The Clock Room and the Macro-Cognitive Accounting Framework: A Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms, a five-volume book (in both Chinese and English editions), is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

License summary: https://creativecommons.org/licenses/by-nc-nd/4.0/deed.en Legal code: https://creativecommons.org/licenses/by-nc-nd/4.0/legalcode.en

Separately, pursuant to Article 4(3) of Directive (EU) 2019/790, the author expressly reserves the right to reproduce and extract this work for the purposes of text and data mining, including text and data mining for the training, fine-tuning, or evaluation of artificial intelligence or machine-learning models. This reservation is intended solely to opt out of the exception or limitation provided for in Article 4(1). It does not affect Article 3 or any other applicable exception or limitation, nor does it reduce the rights granted under the CC license.

Uses that fall outside the scope of the CC license and require permission under applicable law require the author's prior written permission, obtainable at theclockroom@139.com.

When sharing this work under the license, you must provide appropriate attribution and a link to the license, indicate whether modifications were made, and not suggest that the author endorses you or your use. The work may be used only for noncommercial purposes, and adapted material may not be shared.

All rights not expressly granted under the license are retained by the author. Third-party materials in this book, if any, are governed by their original rights statements or licenses and are not deemed to be additionally licensed by the author of this book.

Content Disclaimer

The content of this book is academic research and thought experiments; its analysis is based primarily on historical data and published literature. The author has twelve years of experience in macro strategy research at financial institutions and currently holds no securities investment advisory license. This book does not constitute investment advice, technical implementation advice, or policy advice, nor a recommendation or an offer with respect to any security, financial product, or trading strategy. Readers should exercise independent judgment and bear the risks of their own decisions. Past performance is not indicative of future results.

Chapter 23: A Brief History of Academic Knowledge Production—Publication, Evaluation, and the Growth of Knowledge

Abstract

The growth of science reconstructed indirectly from citation series has advanced in stepwise jumps over more than three centuries; in more recent decades, annual global paper output has expanded overall and become highly concentrated. Publication counts have accordingly become a widely used basis for university rankings, funding allocation, and hiring and promotion, and are widely read by institutions as proof of success. This chapter asks what that reading proves: that a paper has been published, that research has gained recognition, and that knowledge has increased are three judgments that can support one another yet cannot substitute for one another—the power of the publication system lies in organizing the three into a single pipeline, and its risk lies in letting the reading of the first judgment replace the verification owed by the other two. The historical part traces how this machine was built: the correspondence network of the Republic of Letters gave way to the periodically published journal; journal publication became a route to professional recognition; postwar commercial publishing and national evaluation systems bound the three functional lines of knowledge circulation, professional prestige, and commercial profit together; and digital infrastructure embedded exit costs within the research workflow itself. The framework part reviews the load-bearing concepts of the fiber framework and connects them to the promise–metric–judgment closed loop: a promise owes verification; judgment verifies, though transmitting its outputs is not the same as completing verification; and readings are treated in assessment

at a distance as evidence that verification has been completed—evidence that cannot itself deliver verification. The institutional history and evaluation records of academic publishing provide a richly documented calibration case for this loop. The diagnostic part rereads this history in fiber language: publication counts as readings whose fibers are compatible with multiple true states, from major discovery through compliant production to fragmented slicing; the impact factor as pseudo-small fiber; the evaluation system as the carrier of dynamic corrosion; rankings as closed-loop pseudo-calibration. The boundary between functional and pathological recursion lies not in whether the institution reproduces itself, but in whether its reproduction remains answerable to epistemic scrutiny—publication inflation thereby becomes the fuel of rule self-certification, and the acceleration of production by artificial intelligence (AI) amplifies the verification gap. The chapter concludes by arguing for preserving an entry point for judgment: annotate the space of possibilities beside every reading—annotating the fiber—rather than replacing judgment with a cleaner count.

Keywords

academic evaluation; publication inflation; promise; fiber; pseudo-small fiber; dynamic corrosion; pathological recursion; cognitive debt; commensuration; calibration case

Bibliometric analysis based on citation series identifies the growth of science across more than three centuries as a structure of stepwise jumps: from around 1650 to the mid-eighteenth century, annual growth was below 1%; from then until the period between the two world wars, 2% to 3%; and by around 2010, 8% to 9%—each segment roughly three times the previous one. These are growth rates of the citation series, not growth rates of paper output over three centuries. The study's direct comparison of publication and citation growth using more complete data covers only 1980–2012, and it explicitly warns that incomplete early coverage can overestimate growth (Bornmann & Mutz, 2015). The annual output of global science and engineering papers has grown from about 2.2 million in 2014 to about 3.3 million in 2023 (Scopus coverage) (National Science Board, 2025). Publication output has also increased substantially in recent years: in Hanson et al.'s sample of journals indexed in both Scopus (Elsevier's citation database) and Web of Science (Clarivate's citation database), the 2022 article total was approximately 47% higher than in 2016, outpacing the growth in the number of researchers, so that the publication volume a single researcher must handle has risen markedly; Hanson et al. call this condition "the strain on scientific publishing" (Hanson et al., 2024). Output is at the same time highly concentrated: among the roughly 15.15 million named authors indexed in Scopus from 1996 to 2011, fewer than 1% (about 150,000) published continuously every year without interruption; it is precisely this core group that co-authored 41.7% of all papers of the period, and 87.1% of papers cited more than a thousand times (Ioannidis et al., 2014). Alongside the inflation there is one further signal: a preprint analysis based on retraction records from 2022–2024 (including only countries with at least 100 retractions) found a correlation of 0.85 between a country's publication growth over the previous five years and its retraction indicator (retractions per thousand publications, counted by retraction date, with 2024 publication volumes extrapolated—not the final retraction probability of a single publication cohort) (Agrawal, 2025).

The inflated count has not stayed inside the literature: university rankings, funding allocation, and hiring and promotion all read publication counts directly or indirectly and take their growth as proof of success. What this chapter asks is precisely what that reading proves. Three judgments must be distinguished here that can support one another yet cannot substitute for one another: that a paper has been published, that research has gained recognition, and that knowledge has increased. A paper's entry into the public record means it can be read,

discussed, and tracked; whether it has advanced a sound argument, supplied reliable new material, or changed the understanding of a question requires separate discernment. The power of the publication system lies in organizing these three judgments into a single pipeline; its risk lies in letting the reading of the first judgment replace the verification owed by the other two. When the inflated count itself becomes the proof of institutional success, the question is no longer merely one of academic quality: how this machine of inflating output was built, how its readings circulate, and which true states are merged under the same count—these are historical questions and epistemological questions alike.

The answer unfolds along the structure of this chapter. §0 first connects the minimal vocabulary needed to the book's existing framework: promise, metric, and judgment form a functional closed loop—a promise owes verification; judgment is situated and not fully delegable—its evidence, reasons, and conclusions can be transmitted, but transmission itself does not complete verification on the receiving end; and readings are treated in assessment at a distance as evidence that verification has been completed. The institutional history and evaluation records of academic publishing provide a richly documented calibration case for this loop. Part I traces the history of how the machine was built, Part II reviews the load-bearing concepts of the fiber framework, and Part III returns the history to the framework—the diagnostic positions of the impact factor, the evaluation system, and rankings, and how publication inflation becomes the fuel of rule self-certification while the acceleration of production by artificial intelligence (AI) amplifies the verification gap. Part IV is the self-referential examination and the conclusion.

§0 Promise–Metric–Judgment: Terminology Alignment and the Calibration Claim

The role of §0 in this chapter is that of terminology alignment—connecting the new term "promise" into the diagnostic language established over the first 22 chapters of this book, not proposing a new framework parallel to the existing one. All passages that state mechanisms point to the earlier chapters by reference, avoiding a restatement of existing conclusions in another set of words.

§0.1 The Three Terms and the Increment Claim

Promise is an undertaking owed to another party either to bring about a specified state of affairs or to stand behind the reliability of a claimed state of affairs. The scholar's "this research conclusion is reliable" addressed to the academic community, the investment adviser's "this portfolio will deliver risk-adjusted returns" addressed to the client, and the advertiser's "this product deserves your attention" addressed to the audience are all promises. The core feature of a promise is that **it entails an obligation of verification**—making the statement does not establish that the obligation has been fulfilled.

Judgment is the operation that verifies whether a promise has been fulfilled. Its core features are these: judgment addresses a particular question in light of the evidence available and remains open to re-examination and revision; it is situated and not fully delegable—evidence, reasons, and conclusions can be transmitted, but transmission itself does not complete verification on the receiving end. Judgment is the action of verification within the "time dimension" already defined in Chapter 1 §5—the "judgment" side of Weizenbaum's (1976) *From Judgment to Calculation*. What von Neumann (1955) emphasizes is sustained, context-sensitive judgment rather than a once-and-for-all technological solution. This supports the emphasis on situated judgment,

not a definition of its limits of delegation.

Metric is what switches judgment into a reading that can travel across distance, time, and subjects—a reading that represents selected features in a transmissible and comparable form. This chapter examines a particular institutional use of metrics: readings cease to support judgment and are treated as sufficient substitutes for substantive verification. In the book's existing language, the metric is the reading obtained when the mapping function (π), formally defined in Chapter 1 §3.4, acts on the true state—that is, the observable end of the six-variable accounting in Chapter 3 §2.

The three are not three parallel concepts; they form one **functional closed loop**: a promise owes verification; judgment verifies, though transmitting its outputs is not the same as completing verification; a metric travels but cannot substitute for verification. These relations form a feedback loop when verification changes claims, recognition, and subsequent commitments. The loop can become self-certifying when metric attainment is accepted as fulfillment and used to authorize further commitments of the same kind; §8.3 unfolds its institutional operation.

Terminology increment claim. Judgment and metric already have equivalent definitions in Chapter 1 and Chapter 3 of this book; **promise is this chapter's only new term relative to the previous 22 chapters, and its definition is §0.1**. Its locus is the relational structure characterized by Psychosocial Homeostasis (PSH) in Chapter 5 §2—Chapter 5 takes "the interpersonal nexus" as its basic unit, with role and affect as its two lines of connection; "promise" identifies the obligations of fulfillment and verification within that relational structure (the verification it owes).

§0.2 Structural Pressure and Discretization

Once a promise crosses the range of immediate, face-to-face fulfillment (across groups, generations, sovereignties, and scales), judgment no longer suffices—because transmitting judgment's outputs does not complete verification on the receiving end, while the promise at a distance still owes verification. The basic option is to verify by means of transmissible representations—most commonly, **cutting judgment into partial measurements that can travel**, so that those not present can also verify. This book uniformly calls the act of "cutting" **discretization**.

Definition of discretization. Cutting a continuous, holistic, relational object into countable, summable, transmissible units. It differs from discretization in mathematics and signal processing (the sampling of continuous functions)—the discretization of this book is an **operation at the epistemological level**: what is cut is "the wholeness of judgment"; the purpose of the cut is "verifying promises across time and distance"; the cost of the cut is that "the substance of the whole is reduced, in transit, to summable parts." The switch from the "time dimension" to the "clock dimension" in Chapter 1 §5 gives the most basic formulation of this operation; this section distinguishes four forms of the operation.

Discretization has at least four forms in this book (the four forms below can overlap; they are neither mutually exclusive nor exhaustive):

Form	Definition	Typical examples	Main locations in this book
Commensuration	Placing heterogeneous features on a shared measure for comparison (commensuration)—a principal form of metricization	h-index, Sharpe ratio, completion rate, CPM	Chapters 23, 24, 27, and 30
Unitization	Cutting a whole into exchangeable, purchasable independent units	Medieval indulgences (the object of comparison in Chapter 25 §3)	Chapter 25 §3
Time-slicing	Cutting continuous processes or labor into hours/minutes/seconds	Ghost work billed by the hour, customer-service labor billed by the minute, cloud computing billed by the second	Chapter 28 §3
API-fication	Cutting business processes into machine-queryable endpoints	AI agents querying price / promotion / inventory / delivery	Chapter 30 §6

The four forms share one structure: what is cut is a whole; the purpose of the cut is transmission across time and distance; whether judgment exits depends on whether a reading gains exclusive evaluative authority. **Commensuration is the form that collides most directly with "judgment"**—this chapter develops it above all (as the principal form of metricization), because in the academic evaluation scenario this form has a comparatively simple carrier (freer of systemic risk than finance, freer of platform economics than online traffic). Other chapters will use the other forms in their own scenarios, but all point to the same structural operation.

This "cut" is not optional; it is a structural pressure of scaling—once promises scale, substitute verification must borrow some transmissible representation, in evaluation scenarios usually some form of discretization. This explains why three seemingly unrelated domains—academia, finance, and advertising-traffic—have all moved toward discretization: not because they share an ideology, but because they are all doing the same thing—verifying promises across time and distance—and transmissible readings produced by discretization are the most commonly used instrument for it.

§0.3 Closed-Loop Self-Reinforcement: Where the Mechanisms Sit in Earlier Chapters

Once the metric replaces judgment, the closed loop begins to reinforce itself. This dynamic is not a discovery of this chapter but a composite projection, along the "promise–metric" path, of mechanisms already developed in Chapters 1, 3, 4, 10, and 15 of this book. This section gives only positional pointers; the strict formulations of the mechanisms all sit in the chapters cited:

- **The metric rewrites the definition of the promise** (a "good paper" becomes a "highly cited paper")—

this is the direct corollary of the second layer of the three-layer irreversibility in Chapter 1 §4, "dynamic corrosion," acting on the promise side: once readings are used for reward and punishment, actors adjust their behavior around the readings, and the mapping function (π) is rewritten by its own use. See Chapter 1 §4.2; the "publish or perish" (PoP) scenario is developed in §5.3 and §7.3 of this chapter.

- **Fulfillment becomes metric attainment**—this is an instance of the general mechanism of algorithmic *Xiangyuan* (village worthies, or the appearance-only hypocrite) from Chapter 10 under metric governance: the optimized system learns an output structure that conforms outwardly without substantive alignment—an outward-inward discrepancy—fitting the evaluation function on the visible subspace while the hidden subspace is left unconstrained by that objective. See Chapter 10 §§3–5; §7.3 of this chapter concretizes this through the convergence of five traditions—Goodhart, Campbell, Espeland, Scott, and Zhao Dingxin.
- **Meta-metrics calibrate old metrics** (h-index → Q parameter → …)—this is the replay, at the meta level, of "using a clock to calibrate a clock" as critiqued in Chapter 1 §1.3 (the common structure of the first four intuitive strategies); it is also a direct instance of the "meta-clock" in §1.4 and of the closed-loop pseudo-calibration in §3.6 of that chapter, and it connects with the pseudo-calibration state of Chapter 4 §2.4: multiple mapping systems reference one another without tapping an independent calibration channel; the visible subspace is polished ever cleaner while the relative weight of hidden-subspace bias rises. See Chapter 1 §§1.3, 1.4, and 3.6, and Chapter 4 §2.4; the "publish or perish" (PoP) scenario is developed in §7.4 of this chapter.
- **Cognitive debt accumulation**—this is the direct application of shadow debt (D_c^{shadow}) from Chapter 3 §4 together with the "accumulation → concealment → liquidation" three-stage lifecycle (Chapter 3 §4.2); Chapter 15 §§3–5 give three institutional diagnoses of this dynamic in the AI scenario. See Chapter 3 §§4.1–4.2 and Chapter 15 §§3–5; the three-stage mapping of the "publish or perish" (PoP) scenario is given in §7.6 of this chapter.

The four steps are not four parallel diagnoses but one and the same dynamic, which the first 22 chapters of this book have already treated in their own domains. Taking the academic evaluation scenario as its calibration, §7 of this chapter runs the four channels through the same system—the chapter's increment lies not in new mechanisms but in the reconstruction of institutional history and in the synthetic inferences of §0.4.

§0.4 Why Retreat Is Difficult: Algorithmic *Xiangyuan* and Affective Attachment

Closed-loop self-reinforcement does not explain why people do not retreat. Retreat here has a definite referent: withdrawing from the present state, in which metric readings replace judgment, back to the direct verification of promises by judgment. Its difficulty comes from the superposition of two existing mechanisms, one at the system level and one at the subject level:

- **Algorithmic *Xiangyuan* (Chapter 10 §§3–5)** explains the difficulty of retreat at the system level—within the current preference-fitting paradigm and the default commercial relation, improvements in scores cannot distinguish genuine improvement from surface compliance; there is no endogenous brake on the convergence toward *Xiangyuan* (whether constraints beyond the discriminative function can curb it, Chapter 10 explicitly states, must be judged in the concrete deployment context); surface compliance in metrics is therefore a structural tendency, not an individual lapse.
- **Psychosocial Homeostasis (Chapter 5 §2) plus non-human object substitution (Chapter 5 §3)** explains the difficulty of retreat at the subject level—once a metric has long been used to verify promises,

it may enter Layer 3 of the PSH (Intimate Society and Culture; Hsu also lists cherished objects and ideas in this layer) as a controllable non-human object, carrying affective attachment. Giving up the h-index is therefore not changing a tool but giving up part of the self-identity of "I once used it to fulfill academic promises"—the cost of retreat rises with the duration of use. (This is an application of the mechanisms of Chapter 5 §§2–3 to the h-index scenario, not a conclusion Chapter 5 has established; the original text of Chapter 5 uses the conditional mood—if the loop persists, it may compress the human inhabitants of Layer 3.)

Institutional incentives and identity attachment jointly raise the cost of exit: Chapter 10 shows that surface compliance has no endogenous brake, and Chapter 5 shows that retreat must cross affective and identity costs. The structural difficulty of retreat is therefore this chapter's synthetic inference in the academic scenario—the mechanism components come from two existing chapters, while the synthesis and the academic-side application are this chapter's new construction. Chapter 30 §3 unfolds the meta level of this same question in the synthesis of academia, finance, and traffic; this chapter gives only the application to academic evaluation and does not restate the meta level.

§0.5 The Triadic Positions and Terminological Alignment with the Fiber Framework

The ten concepts of Part II of this chapter (§5) need not be reclassified under "promise–metric–judgment"—the grouping already available in Chapter 4 §2 is their natural skeleton; this section only aligns the terminology, to help readers move between the two languages.

Triadic position	Main terms (chapter of definition)	Sections of this chapter
Judgment–metric seam	mapping function (π) (Chapter 1 §3.4), fiber (Chapter 4 §2), ontological dimension reduction (Chapter 1 §4.3; the information-theoretic counterpart is Chapter 4 §1, Proposition 3), time/clock dimension (Chapter 1 §5)	§5.1, §5.2, §5.4, §5.9
Hollowing of the promise	pseudo-small fiber (Chapter 4 §2.4), narrative load (ϕ) (Chapter 3 §2.3, Chapter 8)	§5.6, §5.7
Closed-loop positive feedback	dynamic corrosion (Chapter 1 §4.2), visible/hidden subspace (Chapter 4 §2), cognitive debt/shadow debt (D_c^{shadow}) (Chapter 3 §4.1, Chapter 15 §3)	§5.3, §5.5, §5.8
Framework self-check	the baroque (Chapter 1, Chapter 2)	§5.10

The first order of narrative load (ϕ) (observable switching) sits at the hollowing-of-the-promise position, the second order (accelerated consolidation) at the closed-loop positive-feedback position—the two-order structure already spelled out in §5.7 and shared by Chapter 14 §§2–5 and Chapter 20 §§3.2–§6. The three groups of

derivative diagnostic tools (the three clocks of §7.6 correspond to Chapter 3 §3.2; the cognitive Reynolds number and narrative leverage ratio of §7.8 correspond to Chapter 3 §§5.1–5.2) are all domain instances of existing indicators, not new increments of this chapter.

§0.6 The Calibration Claim

The following discussion uses the history of academic publishing for case-based explication and process examination of the promise–metric–judgment loop. Academic publishing offers a long institutional history with datable turning points and rich material linking publication records to evaluation rules, so several steps of the loop can be traced through textual and process evidence. The process criteria are given in §2.2 and §8.5. The three clocks, the cognitive Reynolds number, and the narrative leverage ratio remain heuristic mappings in this chapter; their proxies, scales, and thresholds have not undergone scenario calibration. The chapter therefore offers an academic case for further examination, not a quantitatively calibrated general model.

The promise–metric–judgment loop and the four forms of discretization remain available to later chapters as needed. Chapter 24 restates the division of labor among judgment, calculation, and readings in a self-contained way; Chapters 25 through 30 may draw on the same alignment in their §0 or restate it within the chapter under the self-containment principle. The strict formulations of the mechanisms remain in Chapters 1 through 15. This chapter aligns the terminology on the promise side and undertakes the case-based explication and process examination described above, without repeating mechanism statements that belong outside this chapter.

Part I: The History and Diagnosis of the Academic Journal System

§1 From the Republic of Letters to the Knowledge Industry

Most of the core features of the academic journal system are far younger than commonly supposed: peer review has been a universal institution for barely half a century; the idea of the impact factor dates to 1955, with the indicator itself built in the early 1960s (Garfield, 2005); the h-index was proposed only in 2005 (Hirsch, 2005). Yet before the journal appeared, the circulation of academic knowledge already ran on a well-functioning network; after its institutionalization, the three functional lines of knowledge circulation, professional prestige, and commercial profit gradually became bound together—what this chapter calls the "binding of the three lines." The historical reconstruction of this binding process is the precondition for understanding the contemporary crisis of academic evaluation, and also the object of the framework diagnosis in later sections.

§1.1 The Pre-Journal Era: The Social Circulation of Knowledge

Before the academic journal appeared, the main channel of circulation for European scholarly knowledge was the correspondence network among scholars—the so-called Republic of Letters (*respublica literaria*). This network had no formal screening mechanism, no peer review, no impact factor; the credibility of knowledge rested on the personal reputation of correspondents and on repeated cross-checking. As Kronick observes in his history of the scientific journal, the scientific journal, as an instrument serving the aims of dissemination and storage, "entered relatively late" in the whole history of scientific communication and constitutes "perhaps only a small segment of all the activities involved" (Kronick, 1976). Scholars exchanged discoveries, reported observations, and contested priority by letter; correspondence was itself both dissemination and verification.

The significance of this historical starting point lies not in nostalgia but in providing a counterfactual reference: the circulation and verification of academic knowledge do not intrinsically depend on the journal as a particular carrier. The journal system was constructed under specific historical conditions—its functionality is not eternal, and neither is its pathology accidental. Burke's long-duration study of knowledge practices likewise suggests that communication, classification, storage, and use each have their own organizational conditions, and that the history of knowledge cannot be written merely as a succession of ideas replacing one another (Burke, 2000, 2012).

§1.2 The Birth of the Journal: The Organization of Records and Priority (1665–1800)

The two periodicals commonly recognized as the earliest scientific journals—France's *Journal des Sçavans* and the Royal Society's *Philosophical Transactions*—were both founded in 1665. Csiszar's research records a frequently overlooked fact: neither counted as part of an "academic institution" at the time of its founding; contemporaries did not even regard the *Philosophical Transactions* as a "journal"; the format, uses, and credit mechanisms of the journal kept mutating throughout its history, and projecting the twentieth-century conception of the journal back onto the early period is precisely the narrative that must be dismantled (Csiszar, 2018). The core function of the early journal was to record authors' priority over ideas and to communicate research findings—"quality control" and "prestige certification" were later accretions.

The history of peer review itself is far shorter than usually imagined, and its early practice must be distinguished from its universal institutionalization. The Royal Society had developed external refereeing practice by the early nineteenth century; Nature made external review a requirement for research papers only in 1973; the secondhand studies Macdonald cites record the date as 1976—where the two accounts differ, the official record and history-of-science research dating it to 1973 should be followed (Baldwin, 2015; Macdonald, 2025).

The establishment of peer review as a universal institution is better described less as the academic community's long-held defense of quality control than as the growing entanglement of knowledge certification, professional rewards, and commercial operations—commercial publishers adopted the existing refereeing procedures in the 1960s and 1970s and sold them as a publishing service. In Fyfe et al.'s formulation: "Historically, the peer review processes on which academic reward and recognition depend had belonged to disciplinary communities, learned societies and university presses; but their adoption by commercial publishers in the 1960s and 1970s means that they are now sold as a key value-added service to the academic community" (Fyfe et al., 2017).

§1.3 The Era of Institutionalization: The Formation of the Functional Circuit (1800–1945)

The nineteenth century was the crucial century of academic institutionalization. Disciplinary boundaries were drawn, the university became the principal site of research, and the doctorate became the entry credential of the academic profession. The nineteenth-century changes Csiszar studies concern precisely the reorganization of authorship, the public record, and intellectual authority (Csiszar, 2018). In this process the journal gradually shifted from "digest of information" to "core institution of knowledge certification and professional prestige accumulation." In his presidential address to the American Sociological Association, Burawoy proposed four types of sociological knowledge—professional, critical, policy, and public:

> In the best of all worlds the flourishing of each type of sociology is a condition for the flourishing of all, but they can just as easily assume pathological forms or become victims of exclusion and

subordination. (Burawoy, 2005)

What this chapter calls **functional recursion** refers to the self-reinforcing circuit that the academic journal system formed over its historical development: research → submission and peer review → formal publication into the public record → citation → reputation → academic positions/resources → more research (preprints and post-publication review are discussed separately and not merged into this chain). This chapter uses "functional" as an analytical criterion: the circuit's output remains answerable to independent epistemic scrutiny. It does not assert that 1800–1945 as a whole was a golden age in which the system mainly served knowledge itself. The effectiveness of peer review has always lacked strong evidence (Jefferson et al., 2007; Smith, 2006), but this limitation does not prevent the relation to scrutiny from serving as an analytical criterion.

Functional recursion rests on four supports: (1) peer review as an instrument of community self-governance—despite its short history and contested effectiveness, in the phase when the academic community held the power of review it did perform a certain screening and improvement (Bornmann, 2011b); (2) the journal as a traceable public record—Burke lists "record" and "storage" among the core operations of knowledge practices (Burke, 2000, 2012); (3) reputation accumulation as a professional incentive—in the phase when the academic community held the standards of evaluation, reputation gave researchers an incentive to invest in long and arduous research (Slaughter & Rhoades, 2004); (4) academic autonomy as a protective layer for the diversity of knowledge—Merton's norms of science (universalism, communality, organized skepticism, and disinterestedness) provided, in principle, a legitimating foundation for academic autonomy (Merton, 1973).

The core criterion of functionality is not the perfection of each link but whether the circuit retains its tie to epistemic scrutiny: whether output that flows back remains subject to independent verification. The unfolding of pathological recursion is given in the three checks of §2.2.

§1.4 The Commercial Turning Point: The Binding of the Three Lines (1945–1990)

Postwar academic publishing underwent a structural turn. Fyfe et al. record the pivotal year 1954: the Royal Society took over its own sales and marketing that year, ended free distribution to universities, and reclassified libraries from "recipients of gratis copies" to "paying customers"—the 1954 review ended the Society's extensive out-of-commerce distribution to institutions across the world. The free list was gone, and by the early 1970s only about 40 institutions remained in an exchange network. University libraries became customers, and sales became the core form of circulation for the Society's journals; by 1987, the sales team reported that the research journals "no longer had any individual subscribers"—all subscriptions came from institutions (Fyfe et al., 2022).

Commercialization was not an external invasion but a repricing of the academic system's own prestige structure. What commercial publishers took over were journals, author relations, and certification procedures that the community already valued: in Britain, Robert Maxwell from the 1960s onward turned academic journals into a money-making machine, breaking the tradition of learned societies and university presses that "hoped only to cover costs" (Macdonald, 2025); the Big Five commercial publishers followed, acquiring large numbers of quality journals operated by societies and university presses—Reed-Elsevier bought Pergamon Press in 1991 (Larivière et al., 2015). By 2013, the top five publishing organizations by WoS-sample output accounted for about 53% of papers in the natural sciences and medicine (the American Chemical Society, a scientific society, among them), and together more than half in the social sciences and humanities (Larivière et al., 2015).

The core change of this stage is the binding of three functional lines: the knowledge-circulation mechanism + the professional-prestige mechanism + the commercial-profit mechanism. Before, the three were connected yet separable; after, publication was no longer only an act of knowledge dissemination but simultaneously a "professional prestige token" and a "source of commercial profit." Fyfe et al. summarize this as the splitting of three roles: the validation of knowledge, symbolic capital, and the profit-making enterprise (Fyfe et al., 2017).

The structural basis of the binding is supplied by two lines of political-economic analysis. Knoche points out that under "publish or perish" pressure, scholars hand their finished work to publishers free of charge, as a gift:

> The handover of academic products to capitalist publishers is usually even done free of charge for the publishers, i.e. as a kind of gift, although the products are manufactured by wage or salary earners paid by the state using publicly owned means of production. (Knoche, 2025)

The substance of this "voluntary" delivery is the pressure of the qualification regime—scholars are formally voluntary but in fact under the pressure of the prevailing academic qualification system; and what libraries face is a guaranteed market in which they are "under an obligation to purchase": the foundation of publishers' extremely high prices and super-high profits is precisely this market combination—"finished products delivered free of charge under publication pressure" plus "libraries guaranteed to purchase under a clear procurement obligation" (Knoche, 2025). Posner adds the law-and-economics perspective: even without copyright, scholars would continue to publish—both because of the "publish or perish" regime and because of the significant non-pecuniary income that publication brings; the loss of copyright income would have almost no effect on the volume of academic publication (Posner, 2014). As far as academic publication is concerned, copyright income is not the principal incentive for producing these papers; the distribution of returns from copyright control also depends on ownership, licensing, and publishers' pricing power, and cannot be inferred solely from authors' continued willingness to publish—but the structural fact that the publishing end controls the rights, combined with the libraries' procurement obligation, already suffices to support this section's political-economic judgment that the intermediary benefits.

§1.5 The Digital Era: Concentration Rather than Dispersion (1990 to the Present)

A widely held expectation was that digital technology would weaken the publishers' intermediary position. The empirical outcome was the opposite. Larivière et al. found that, among the natural and medical science papers they analyzed, the five largest publishers' combined share rose from just over 20% in 1973 to about 53% in 2013 (Larivière et al., 2015). From the late 1990s, publishers converted single-journal subscriptions into journal bundles—the so-called Big Deal: selling libraries online access to all of a publisher's journals at a price equivalent to the total of existing print subscriptions; this model soon placed university libraries in an "all or nothing" position; according to the ARL statistics cited by Gagnon, serial expenditures at the surveyed research libraries rose by 402% between 1986 and 2011 (Gagnon, 2017)—a figure that reflects expenditure growth, not the price increase of a fixed set of journal bundles. The European University Association (EUA), representing 800 European universities, submitted a report to the European Commission's Directorate-General for Competition systematically listing eight problems of the academic publishing market—highly concentrated ownership, oligopolistic structure, opaque pricing, high profits resting on public funds, open-access appeals not lowering prices, asymmetric negotiating power, a trend toward vendor lock-in, and no sign of improvement in competition (EUA, 2018); Elsevier, by acquiring the submission system (Aries) and reference management (Mendeley), and by developing the evaluation tool (SciVal), extended its operations from submission and reference management

to research assessment.

Publication output and its growth were concentrated among a small number of publishers. In the jointly indexed journal sample described in the opening, the article total grew by about 47% between 2016 and 2022, outpacing the growth in the number of researchers (Hanson et al., 2024); over the same period, more than 70% of the increment concentrated in five publishers—by increment: MDPI (about 27%), Elsevier (about 16%), Frontiers (about 11%), Springer (about 9.5%), Wiley (about 7.0%)—the five largest contributors to publication growth reshuffled the incumbent five, with Elsevier, the largest publisher by publication volume, falling behind MDPI in the increment (Hanson et al., 2024).

The open access (OA) movement originally sought to break the commercial lock-in, but after the implementation of gold OA (the immediately accessible model on the publisher side, a substantial share of which charges authors an article processing charge), the article processing charge (APC) became a new profit stream: in APC-charging open models, publishers commonly adopted pricing strategies that maximize "volume × APC," making APC revenue a significant pillar of their profits (Peng et al., 2025). Publishers' charging targets extended from libraries to authors and funders; fully open access, hybrid publishing, and read-and-publish agreements have different charging structures and should not all be described as charging twice for the same article. The shift from reader-pays to author-pays can remove access barriers, and can equally form new publication thresholds—rising APC expenditure shows that publication fees absorb research resources; assessing exclusion requires evidence about who bears those fees, available waivers, and work that could not be published (Peng et al., 2025).

The aggregate figures conceal a key dimension: the attribution of growth—the estimates of Ioannidis et al. cited in the opening (the continuously publishing group, which accounted for fewer than 1% of all authors, co-authored 41.7% of all papers and 87.1% of the most-cited ones; the full set of figures is given in the opening); this paragraph discusses only its implications for the attribution of growth (Ioannidis et al., 2014). The growth of total publication accompanied the expansion of the researcher population and the continuous participation of the core group; these figures cannot by themselves decompose the respective contributions of population, collaboration size, and individual output, nor can growth be attributed to any one of them—what they show is that continuous publishers participated in a large proportion of paper production, and that growth in totals occurred together with population expansion and the continued participation of the core group.

Preprints: a parallel second track. Digitization did not serve only commercial concentration; it also gave rise to dissemination channels bypassing journals. arXiv was created by Paul Ginsparg in 1991, serving physics, mathematics, and computer science; bioRxiv was launched in 2013 by Cold Spring Harbor Laboratory, bringing a similar arrangement to biology (Sever et al., 2019). Taking the local arXiv archive on which this book's writing is based (grouped by the year of first submission encoded in the paper identifier[1]): 278 papers held in the archive for 1991 (the arXiv platform began operating in August of that year), about 241,000 in 2024, about 255,000 in 2025, and about 212,000 in the first nine months of 2026—2024 averaged about 20,000 per month, and the figure is still climbing (about 23,500 per month in the first nine months of 2026). But arXiv's coverage has long differed greatly across disciplines: the early studies compiled by Larivière et al. show that in high-energy physics almost all journal papers passed through arXiv first (2009 study), in mathematics about 18.5%

[1]Statistical note: the figures are deduplicated by the arXiv identifier carried in the filename from the local arXiv paper archive (1991 to September 2026) on which this book's writing is based, totaling about 2.33 million papers; downloads for 2014 and the 2018–2019 interval are incomplete, and only figures from fully covered years are used here. This note reports only the holdings of the local collection; its completeness has not been checked year by year against the official arXiv catalog; cross-year monthly comparisons use only full operating years.

(2007 study); among WoS-indexed papers of 2010, only 3.6% across all disciplines simultaneously appeared on arXiv (Larivière et al., 2014)—these proportions describe an early distribution and must not be taken as current coverage.

The core institutional meaning of the preprint is the decoupling of dissemination from evaluation—researchers can share work immediately and establish priority without waiting for peer review and journal certification. This returns to the social circulation of knowledge in the "Republic of Letters" described in §1.1. Harnad foresaw this trend as early as 1990, proposing the concept of "scholarly skywriting"—bringing prepublication communication, from informal discussion through preprints to formal publication, entirely into the online environment (Harnad, 1990). He directly inverted the "publish or perish" formula; under the title "Publish or Perish—Self-Archive to Flourish," he advocated the green open access path of self-archiving papers in institutional repositories, making research results freely visible to all (Harnad, 2006). Berners-Lee et al., discussing web intellectual property, cited this scheme to support the judgment that the free reuse of academic information is harmless and desirable (Berners-Lee et al., 2006).

But the preprint is not an antidote to "publish or perish." About 70% of bioRxiv preprints were published in traditional journals within two years—a proportion that is still an underestimate, since changes of title or author may escape the matching algorithms (Sever et al., 2019)—this proportion indicates coexistence rather than substitution within the sample; preprints do not automatically change hiring and promotion rules, but they can enable earlier scrutiny. Career advancement still takes journal publication as its standard: the preprint decoupled dissemination from evaluation, but not the incentive structure of "publish or perish." The technical means bypassed the commercial publishers' intermediary position without bypassing the deeper discretizing operation of "publication count = academic value."

The most counterintuitive result of the digital stage is thereby complete: technological progress did not weaken the intermediary, but served stronger market concentration; digitization and open access did not automatically release that dependence, and the concrete changes still vary by discipline, publishing model, and institutional arrangement; in APC-charging models, the party being charged shifted from readers and libraries to authors and funders.

§2 Pathological Recursion: When the Circuit's Output Points to the System Itself

§2.1 The Rise of Audit Culture

Shore and Wright define "audit culture" as the situation in which "the principles, techniques and rationale of financial accounting have become dominant features of the way society is organised"—it is not a type of society standing alongside "feudal society" or "capitalist society," but a rationality of governance and the corresponding dispositions and practices (Shore & Wright, 2024). Power, in *The Audit Society*, calls audit a "ritual of verification," and delivers the sharpest judgment of the whole book:

> The audit society is a society that endangers itself because it invests too heavily in shallow rituals of verification at the expense of other forms of organizational intelligence. (Power, 1997)

Shore and Wright add the paradoxical effect of the rituals: these new tools, set up for good governance, transparency, and credible reporting, often produce their opposite—poor governance, opaque decision-making, and growing cynicism and distrust (Shore & Wright, 2024; their paraphrase of Power).

Audit techniques diffused from the British education sector of the 1980s into health care, municipal administration, and the military, then flowed back into the university. From the Jarratt Report of 1985, university management began to be corporatized: vice chancellors became chief executive officers, personnel departments became "human resource management," departments became "cost centers," academics became "units of resource" (Shore & Wright, 2024). The RAE (Research Assessment Exercise) evolved step by step from seven-point marking into a mechanism of resource concentration: by 2001, 75% of research resources were concentrated in the top tier of departments (Shore & Wright, 2024). They point out that audit is not a neutral technical instrument but brings with it the process of "domaining"—a comprehensive reshaping of the manner of management and governance of individuals, organizations, and even states; audit was extended from its original anchorage in financial accountability into a general tool of governance, colonizing ever more domains of everyday life (Shore & Wright, 2024). This analysis supplies the institutional-level explanation for understanding pathological recursion: pathological recursion is not merely a problem internal to the academic community, but a structural consequence of the transformation of the rationality of state governance.

§2.2 The Reversal from Functional to Pathological

This chapter uses **pathological recursion** for an operating state in which the above relation to scrutiny is weakened: internal output increasingly serves institutional reproduction, metric proliferation, or commercial profit while independent scrutiny is eroded. This definition does not presuppose a single historical reversal that has already been completed. Pathological recursion is not the simple collapse of functional recursion, but its distorted continuation after being "taken over"—the same circuit structure, with the direction of the incentives reversed.

Gingras states the mark of this reversal precisely: the scholarly world formed a false equation—the "sudden discovery" of bibliometric data produced "the false and unfortunate equation that bibliometrics = evaluation, as if it had no other uses outside the evaluation game" (Gingras, 2016). The root of commensuration was the ideology of "new public management" of the 1980s—the state, university managements, and review systems increasingly wanted to translate research activity into comparable, rankable, accountable numbers. Gordon describes the compression process: the perceived quality of a journal's content was reduced to the journal's reputation, and the journal's reputation was further reduced to a quantifiable equivalent—a mode of knowledge production was thereby shaped that served a single master rather than remaining a critical interlocutor of the discipline (international law) to which it belonged (Gordon, 2017). When the scientific paper changed from "unit of knowledge" into "accounting unit" (Gingras, 2020), the circuit's output turned from pointing toward knowledge to pointing toward the reproduction of the institution itself.

This distinction requires checking at least three connected things. First, what later judgments actually depend on: material and inference, or publication venue, citation counts, and prior recognition. Second, how these grounds return: does an added round of processing add examination, or does it package the same source as multiple confirmations. Third, how this return changes the conditions of verification in the next round: does it preserve room for counter-evidence, the formation of material, and professional judgment, or does it make these activities harder. The three checks revolve around one question—what relation the circuit retains to the object of the knowledge it claims—not three isolated new definitions.

Whether epistemic success and failure can change recognition, resources, and research choices is an important checkpoint of the above processes at the distributive end of the institution, not the whole meaning of pathological recursion. A false claim, gaining stronger authority through mutual relay, may constitute pathological feedback

even if funding relations are unchanged; a project defunded for failure does not show that the institution has protected independent judgment, for the failure may itself be the knowledge yielded by serious examination. Correspondingly, functional feedback should accommodate negative results and the clarification of boundaries, not reward success stories only.

§2.3 Mechanisms of Pathological Recursion

Usurpation by the indicator. Goodhart, from monetary policy, and Campbell, from social assessment, each proposed a related judgment: when outcome measures enter control and reward and actors can adjust around them, strategic adjustment or selective adaptation may change both activity and the relation between the measure and its goal (Goodhart, 1975/1984; Campbell, 1979). Campbell's original formulation deserves quotation in full:

> The more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor. (Campbell, 1979)

Strathern supplied the restatement that has since been widely quoted—she herself notes that it states exactly Goodhart's law:

> When a measure becomes a target, it ceases to be a good measure. (Strathern, 1997)

In the academic scenario, Bornmann conceptualizes this operation as "mimicry in science": like certain animals adapting to their environment to evade predators, scientists adapt their research behavior to bibliometric indicators. Four adaptive strategies are identified: following the mainstream of one's field and avoiding unconventional research (risky, interdisciplinary, long-term); pursuing short-term research for faster publishable output; submitting to low-quality but bibliographically indexed journals; and slicing research findings into salami-thin pieces published separately (Bornmann, 2011a).

Governance by rankings. Espeland and Sauder argue with the concept of "reactivity": people change their behavior because they are being evaluated, observed, measured; two mechanisms—the self-fulfilling prophecy and commensuration—let rankings reconstitute the social world they measure (Espeland & Sauder, 2007). Uslu, using a sample of 224 universities in the 2018 editions of four ranking systems (ARWU, THE, QS, URAP), estimates in a regression analysis with reorganized indicators that research-reputation components (including publication, citations, and reputation) together contribute 73.71% of university ranking scores (Uslu, 2020)—a result for that sample and model, not an official weighting adopted uniformly by the rankings; rankings are constituted mainly by research-performance indicators, of which the part attributable to historical accumulation is analyzed below by Stack. Stack points out that 30% of the weighting of the ARWU (Academic Ranking of World Universities, commonly known as the Shanghai Ranking) is based on the number of Nobel Prize and Fields Medal winners among an institution's faculty or alumni, and that ranking advisories, the leadership of top-ranked institutions, and Nobel adjudication committees form an "echo chamber" (Stack, 2020).

The integrity crisis. Baccini et al. propose the "inwardness" indicator—the share of a country's citations that are self-citations—measuring collective strategic citation at the national level: the Italian scientific community's pervasive strategic use of citations takes two forms, strategic author self-citation and "citation clubs" (Baccini et al., 2019). Around Italy's university reforms of 2010 and the assessment rules that followed (ANVUR's legal basis dates to 2006 legislation; its organization and operation were specified by Presidential Decree No. 76 of

2010), inwardness rose from 20.62% in 2000 to 30.73% in 2016, an increase of 10.11 percentage points over the whole period, the largest among the G10; of this, 8.29 percentage points were added between 2008 and 2016, more than 4 points above the G10 average, and 23 of 27 Scopus subject categories showed the highest inwardness growth (Baccini et al., 2019). Catanzaro reports the "citation cartel" phenomenon—institutions in China, Saudi Arabia, and Egypt mass-producing low-quality papers that cite one another to inflate citation counts—for which Clarivate in November 2023 excluded the entire field of mathematics from its list of Highly Cited Researchers (Catanzaro, 2024). Alongside the inflation there is also the retraction signal—already stated in the opening: the correlation between a country's publication growth over five years and its retraction indicator reaches 0.85 (Agrawal, 2025). Predatory journals, as a direct product of "publish or perish," are characterized by the consensus definition of Grudniewicz et al. as "entities that prioritize self-interest over scholarship," marked by false or misleading information, deviation from best editorial and publishing practice, lack of transparency, and aggressive and indiscriminate solicitation practices; as long as universities continue to use publication counts as criteria for graduation or promotion, "this threat is unlikely to disappear" (Grudniewicz et al., 2019).

The accumulation of cognitive debt. Redner positions publication as a mechanism of knowledge certification rather than mere information dissemination:

> Publication does not just record and communicate knowledge; even more crucially it certifies it as knowledge and not just mere information, conjecture, opinion, belief or anything else short of knowledge. (Redner, 2023)

When this oversight is missing or corrupted, "science founders in a sea of unreliable publications"; he lists as typical presentations of the "publish or perish" syndrome: young scholars publishing their dissertations, senior scholars publishing their lecture notes, and the oldest and most famous scholars publishing their memoirs (Redner, 2023). Hendricks reconceptualizes the predicament as a coordination failure under non-common knowledge: individual critical awareness does not by itself produce common knowledge—knowing that the system is distorted does not mean knowing that others know, still less knowing that others know that you know; those who continue to act as if the norms were still shared are "suckers," those who recognize that the assumption no longer holds and exploit this asymmetry are "hustlers" (Hendricks, 2026). Beigel et al. summarize the systemic cost as a "fourfold drain"—the money, time, trust, and control of the research system being continuously drawn off: the four major publishers generated more than $7.1 billion in journal revenue alone in 2024, with cumulative profits of more than $14 billion from 2019 to 2024, Elsevier's profit margin staying above 37% for five years, while article processing charges brought the leading publishers nearly $9 billion in revenue between 2019 and 2023—according to the estimate of Haustein et al. (as cited in Beigel et al., 2025).

§3 A Multidimensional Diagnosis of the Evaluation Crisis

The evaluation crisis is not the discovery of a single discipline. Scholars of different traditions enter the problem from their own perspectives, each capturing a different facet of the same structural problem; the complementarities and tensions among the facets are themselves diagnostically valuable. This section is organized not as a list of "perspectives" but by the analytical questions the traditions are asking—questions are the true interface of the dialogue.

§3.1 The Debate over the Turning Point: Was There a Functional Golden Age?

§1 and §2 have recorded three inflection points: 1954, when the Royal Society reclassified libraries from recipients of gratis copies to paying customers (§1.4); 1985, when the British Jarratt Report corporatized university management and audit techniques were unmoored from financial accountability into tools of governance (§2.1); and the 1990s, when the scholarly world formed the false equation "bibliometrics = evaluation" (§2.2). The three inflection points reveal changes in constraining conditions, but they do not by themselves prove a linear decline from a functional golden age to a pathological era; "functionality" must be judged case by case through the relation to scrutiny specified in §1.3.

Csiszar raises a fundamental challenge to this presupposition. His point is not that some inflection point has been misdated, but that the popular framework which diagnoses the current crisis as a mismatch between scientific reward structures and the true goals of scientific research is itself a Cold War legacy—Cold War observers and sociologists described science as an autonomous, self-regulating social system with the scientific literature and its gatekeepers at the core; the current crisis narrative (unintended consequences, Goodhart's law, quantification distorting research agendas) all rests on this image (Csiszar, 2018). Csiszar further notes that to assume the epistemic virtues of scientific life ever developed in isolation from economic interests, technological visions, or political commitments is a naïve assumption. If Csiszar is right, the "functional golden age" never existed in pure form—pathological recursion is not a "corruption" of functional recursion but the direction of tendency of indicator governance in the absence of constraints; the question is rewritten from "how to return to the past" to "which conditions once restrained it, and why they failed." Here, "not a corruption" means only that the historical starting point must not be imagined as a pure golden age; it does not deny that a functional circuit can undergo a reversal of direction.

This judgment shifts the baseline of the diagnosis: not "how to return to the past," but "what were the buffers that once existed, and why did they fail." §3.7 returns to this question.

§3.2 The Debate over Mechanism: When Metrics Become Targets, How Does Gaming Become the Norm?

§2.3 has already unfolded Goodhart's law and Campbell's principle: when outcome measures enter control and reward and actors can adjust around them, strategic adjustment or selective adaptation may change both activity and the relation between the measure and its goal; Strathern's restatement—when a measure becomes a target, it ceases to be a good measure—has since been widely quoted. This section adds the two concrete forms this mechanism takes in the academic scenario.

The first form is the institutionalization of gaming. Macdonald records:

> Gaming is now embedded in academic performance measurement: academics who do not game rapidly become uncompetitive and perish. Producing indicators of performance is easier than producing the performance itself, and is also less intellectually onerous. (Macdonald, 2025)

The individual-level confessions are more direct. One scholar, speaking of one of his own papers:

> From an instrumental position, [the paper] gave me my professorship. But as a scholarship piece, it's disgusting. Yeah, it was really, really awful. (author quoted in Butler and Spoelstra, 2014; as cited in Macdonald, 2025)

Another, more bluntly:

> … it doesn't matter what the fuck you publish, only where you publish it. (academic quoted in Butler and Spoelstra, 2014; as cited in Macdonald, 2025)

The importance of these two self-reports lies not in passing judgment on all papers, but in the researchers' simultaneously holding two evaluations: they know how the work helped their careers, and they do not equate that help with a contribution to knowledge.

The second form is the camouflage function of quantitative language. Bergstrom and West use the Hanoi rat-tail bounty as the illustrative case: in early twentieth-century French Hanoi, a bounty was paid per rat tail to exterminate rats, so residents cut off the tails and released the rats back into the sewers to keep breeding, imported rats from other cities, and even farmed them—the bounty program failed because people gamed the system; faced with rewards, people will inevitably do so (Bergstrom & West, 2020). They also propose the concept of "new-school bullshit"—"New-school bullshit uses the language of math and science and statistics to create the impression of rigor and accuracy" (Bergstrom & West, 2020)—and explain its proliferation with three principles: producing bullshit takes less work than cleaning it up, demands less intelligence, and travels faster; the first of these is Brandolini's principle—refuting bullshit requires an order of magnitude more energy than producing it. When academic evaluation goes by quantitative signals, and the quantitative signals can themselves be manufactured at low cost, gaming turns from individual transgression into rational strategy.

§3.3 The Debate over Structure: Individual Moral Failure or Institutionalized Corruption?

When large numbers of scholars simultaneously commit acts that damage the quality of knowledge, is this widespread individual misconduct, or is the incentive structure at work?

Sowell's (1980) analysis preserves the inner tension of this question. On the one hand, the function of "publish or perish" policy is not publication itself, but to force faculty to sort and label themselves by displaying professional competence before peers—what is rewarded is not publication but the ranking of academic ability that publication promotes; he even concedes that a string of mediocre publications may be harmful to the individual yet valuable to the profession in ranking and labeling its members. On the other hand, scholars of real ability who are unwilling to publish much may be "undervalued" by this system—the sorting mechanism works, yet not necessarily by the academic goals worth preserving. An institution can run well at sorting while crowding out the other purposes of research and teaching.

Hendricks (2026) reconceptualizes the predicament as a coordination failure under non-common knowledge (his "sucker/hustler" distinction and full formulation are given in §2.3). His analysis of the attention economy also suggests that when visibility itself becomes the circulating good, headline optimization and novelty inflation are no longer vanity but necessity—this extension is this chapter's reinterpretation in the setting of academic publication; the concept of attention labor in Hendricks's original appears in the context of angry engagement.

Collyer's (2019) interviews with forty-five senior scholars in health-inequality research provide the most vivid empirical material:

> Many interviewees felt that an abundance of mediocre papers is 'polluting the scientific air' and that the question 'is this publishable?' has supplanted 'is this good science?' (Collyer, 2019)

The interdisciplinary barriers she records are especially critical: publication in the economics "top five" is almost a precondition for tenure, while publication in high-impact medical journals such as *The Lancet* does

not help junior economists seeking promotion—the incentive for interdisciplinary collaboration is minimal. Publishability thereby enters the choice of topic before research begins, rather than performing its filtering function after results are complete. Collyer's own publishing choice illustrates the structure she diagnoses: she chose to publish briefly under the *Nature* brand (the comment in fact appeared in *Nature Human Behaviour*) rather than write a full analysis. She writes: "I have made a calculation here that brevity is offset by that magic word, Nature, on my CV." A scholar studying publication pressure, in publishing that very study, obeyed the logic she diagnosed.

Tiles and Oberdiek give the philosophical formulation of this structural debate. Borrowing MacIntyre's theory of practice—a practice being a complex, demanding activity with standards of excellence and internal and external goods—they write:

> Sometimes external goods overwhelm the internal goods of a practice leading to its decay. The need to 'publish or perish', some argue, has led to the corruption of scholarly practice. Practices, in short, never run on 'automatic pilot', at least not for very long. (Tiles & Oberdiek, 1995)

When external goods (publication counts, citations, rankings) overwhelm internal goods (quality of knowledge, depth of understanding), what is damaged is not the individual paper but the legitimacy of scholarly practice itself as a mode of knowledge production. Redner (2023) adds to this diagnosis from the angle of knowledge certification (his certification statement and the canonical list of dissertations, lecture notes, and memoirs are given in §2.3): when publication cannot effectively distinguish knowledge from other information, publication degenerates from an instrument of knowledge contribution into a means of professional survival.

§3.4 The Debate over Deepening: Manipulation Escalates from Content to Signals

Biagioli and Lippman record the transfer in the form of manipulation:

> "Publish or perish" is merging with "impact or perish." (Biagioli & Lippman, 2020)

Traditional misconduct fabricates false knowledge claims (false data, false results); the new manipulation fabricates false signals of impact (false citations, false reviews, false impact factors)—they call this the migration from "epistemic crimes" to "bureaucratic crimes"—post-production manipulation can leave the epistemological content of the paper entirely untouched: a paper whose content is sound may nonetheless acquire a distorted evaluation of impact through fabricated peer review (Biagioli & Lippman, 2020). This means pathological recursion no longer merely pollutes the record of knowledge; it begins to pollute the evaluative infrastructure of the record of knowledge.

National-level evidence comes from Baccini et al.'s inwardness research (the Italian 2000–2016 data were cited in §2.3): during the same period in which Italian inwardness rose sharply, Italy's international collaboration, though long in the lower ranks of Europe, showed no growth commensurate with the rise in inwardness—the authors conclude that expanding international collaboration cannot explain the change, pointing to collective strategic citation induced by the evaluation reform (Baccini et al., 2019).

The most systematic symptom comes from reproducibility research. Several preclinical replication efforts summarized by Begley and Ioannidis reported high failure rates: roughly 75–80% of 67 studies in one project and about 90% of 53 oncology studies in another—findings that indicate serious risks but do not provide a single population-wide estimate for all preclinical research (Begley & Ioannidis, 2015). The Open Science Collabora-

tion's (OSC) replication of 100 experimental and correlational studies published in three psychology journals found: only 36% of replications achieved statistical significance, with replication effect sizes on average about half the originals; 47% of original effect sizes fell within the 95% confidence interval of the replication effect size, and 39% were judged by subjective assessment to have replicated successfully (OSC, 2015). Park et al. analyzed 25 million papers indexed in the Web of Science from 1945 to 2010 and 3.9 million patents from the US Patent and Trademark Office from 1976 to 2010, finding an across-the-board decline in the disruptiveness scores of papers and patents—measured by the CD5 disruptiveness indicator (which, on the basis of subsequent citation structure, measures whether a work opens a new direction or consolidates an existing one), the decline ranges from 91.9% in the social sciences to 100% in the physical sciences; the finding was then replicated on four datasets totaling 20 million papers—JSTOR, the American Physical Society archive, the Microsoft Academic Graph, and PubMed (Park et al., 2023). The result and its sensitivity to data processing remain contested: in 2026, Holst et al. argued that part of the decline can be explained by dataset artefacts such as entries with zero references—removing entries with a CD5 disruptiveness score of 1 shrinks the decline substantially; Park et al. replied that even using the dataset, metric, and method recommended by the critics, the decline for papers and patents remains significant, and that the critics' own data contain a higher proportion of works with zero references (Holst et al., 2026; Park et al., 2026).

These readings must not be read as a single curve: publication growth (stated in the opening), cross-country differences in retraction indicators (the 0.85 correlation of the opening), and declining disruptiveness may share some institutional drivers, but their gauges differ—expanding database coverage, growing co-authorship, and changes in researcher numbers all affect the totals; being uncited is not the same as being valueless, and being frequently cited is not the same as being confirmed. Replacing one count with another count cannot automatically end the disputes of evaluation.

§3.5 The Debate over Domains: Do Different Disciplines Face the Same Problem?

The diagnoses above presuppose a premise not yet questioned: does pathological recursion operate in the same way across all disciplines? Zhao Dingxin's (2019) two temporalities, starting from the structure of social power, supply a deep language for this difference: economic and military competition has clear winning and losing criteria and continuous selection pressure, driving accumulative development time—directional and accumulative; ideological competition is "like a group of Christians, Buddhists, Muslims, and Daoists debating with one another over whose doctrine is superior"—"none can persuade the others"—with no clear winning and losing criteria, driving diversity time—directionless, highly plural, slow (Zhao, 2019; author's translation). This chapter uses these two ideal types as analytical reference points; it does not equate Zhao's economic, military, and ideological categories with disciplinary categories. The disciplinary comparison that follows is this chapter's own reorganization and must be supported separately by evidence from each domain.

In formalized research with explicit rules, such as some mathematical proofs, derivations can be checked within those rules. This does not mean that the verification mechanisms of "theoretical science" as a whole are internally closed, nor does theoretical work in empirical disciplines thereby acquire the same barrier. But within the barrier, distortions of direction and form occur all the same. Mandelbrot records the advice the mathematician Kac gave him:

> Most active young scientists know they must publish articles or perish. But your case is different. Unless you stop this avalanche of individual articles and write a book, I shall let you perish.

(Mandelbrot, 2012)

The sentence's original meaning is one friend's personal advice to an author whose papers were piling up—Mandelbrot's papers were being held back by journals over questions of style, and Kac urged him to break out with a book; reading it as an instance of "publish or perish cutting knowledge into countable fragments" is this chapter's reinterpretation. Distortion of direction has more direct evidence: Wu et al.'s analysis of papers, patents, and software shows that as team size increases from 1 to 50, the disruptiveness percentile falls by 70, 30, and 50 percentile points for papers, patents, and software, respectively. Single authors and five-person teams have similar chances of producing highly cited research, but single-author papers are 72% more likely to be highly disruptive; ten-person teams are 50% more likely to produce high-impact papers and are much more likely to develop existing ideas (Wu et al., 2019). This result indicates a difference between team size and the type of output; whether it translates into a survival disadvantage for small teams through evaluation and funding mechanisms requires separate tracing. And the walled garden itself is being eroded: §2.3 has described the mathematics citation cartels and Clarivate's 2023 exclusion of the entire field of mathematics from the Highly Cited list—internal verification within formal systems cannot stop the post-production manipulation of citation signals.

Empirical science cannot complete verification within a formal system and also requires external replication. This verification link provides an entry point for publication incentives, but the material in this chapter is insufficient to declare empirical science as a whole the "most vulnerable" domain. In the animal-stroke literature compiled by Begley and Ioannidis, negative results account for only about 2% (Begley & Ioannidis, 2015)—the literature of this field is systematically biased toward positive findings. Stroebe et al. point out that "failed replications are not typically published and thus do not become widely known to the scientific community"—science's self-correction depends on correction after replication failure, and the publication system blocks precisely this link (Stroebe et al., 2012). Stern et al. quantify the downstream cost of the corrosion: the Wakefield vaccine–autism paper fueled a measles outbreak in Wales with more than 1,200 cases, at an estimated cost of about £470,000 (about $800,000) (Stern et al., 2014). In medicine, irreproducible research remains in the literature, continuing to be cited and applied clinically—the pseudo-small fiber carries concrete public-health and economic costs.

The social sciences face a problem one layer deeper: not only is the set of true states high-dimensional and the set of observations low-dimensional, the mapping function itself is contested—different theoretical traditions may disagree over "what counts as good research." Shiller gives an instance of topic distortion from macroeconomics:

> Academic research conducted by individuals, who are under pressure to "publish or perish," is unlikely to start data-collection efforts that will help us understand the relatively rare, but serious, depressions and financial crises that occur from decade to decade, but perhaps no more than twice in a lifetime. (Shiller, 2019)

Roncaglia confirms the same mechanism from inside economics: game theory requires rules to be given with absolute precision, and results change with the smallest variation of the rules; multiple solutions and arbitrary variations of rules give the field a vast space of applications (and publications), "this, in the current situation of academic competition ('publish or perish') greatly favours this field of research"—but the result is extreme fragmentation: theoretical fragments can multiply without limit, while falling far short of a general theoreti-

cal understanding of economic behavior (Roncaglia, 2019). In addition, the international index system itself compresses the set of observations: the Web of Science's missing journal coverage and low counts for the humanities (Almeida et al., 2018) leave humanities and social-science scholars without adequate coverage in the international evaluation system over the long run.

In the framework's language, these domain differences can be stated as follows. Formalized research permits rule-governed checking of derivations, although complex proofs may take substantial time and there is no uniform winning criterion for topic choice, importance, or readability. Empirical research relies on observation, experiment, and comparison of conditions; replication requires coordination across laboratories and conditions, but this does not establish that empirical science as a whole must be the most vulnerable. Social research often additionally involves disputes over explanatory frameworks and evaluative aims. These features can coexist within one discipline. Whether an indicator is applicable must be judged by the particular evaluative task and conditions of verification; neither universal cross-disciplinary applicability nor natural suitability for highly formalized disciplines should be presumed.

§3.6 The Epistemological Debate: What Has the Publication System Done to Knowledge Itself?

McGilchrist approaches from cognitive modes: science advances by the perception of patterns, the use of metaphor and formal analogy—Darwin for biology, Freud for psychology, Bohr for physics as the highest examples—and pattern recognition depends on the right hemisphere's wide-field processing; he diagnoses contemporary culture as suffering an imbalance in which the left hemisphere's narrow-focus, abstracting, quantifying mode dominates excessively (McGilchrist, 2021). When academic evaluation systematically favors decomposable, comparable outputs, knowledge production slides from understanding toward measurement—this connection is this chapter's analysis; what McGilchrist himself supplies is the theory of imbalance at the level of cognitive modes. He also has a more direct historical observation:

> Even into the modern era, three of its intellectual giants, whose influence ranges well outside of any speciality – Darwin, Freud and Einstein – did the work for which they are mainly remembered alone, and often without the support (or the straitjacket, depending on how you look at it) of conventional academic institutions. It is worth noting also that, like other intellectual giants such as Russell, Whitehead and Wittgenstein, much (indeed, with the exception of Einstein, almost all) of their influence comes, not through papers, but through lectures or books. (McGilchrist, 2021)

The same passage goes on to note: today's professional scientists do not write books; books carry no career value, because books have no "impact factor" attached to them. When the publication system strips the medium best suited to disseminating groundbreaking ideas of institutional reward, what it excludes is not just a form of publishing but a type of knowledge.

Zhao Dingxin's (2022) theory of the hermeneutic circle supplies the diagnosis at the epistemological level: materials obtained through documents, archaeology, observation, interviews, sampling, and big data are always local information for the case analyzed; the concepts, narratives, and theories built on them are only distillates based on this information—the hermeneutic circle cannot be eliminated, only narrowed; and the "illusion of closure"—the true believer's mistaken sense that the theory has completely explained the phenomenon—is the epistemological root of pseudo-certainty. What commensurated evaluation encourages is precisely the operationalization of the illusion of closure: a number claims to have completed the verification of a judgment, while

the judgment itself never took place.

§3.7 The Counterfactual Debate: Can It Be Reversed?

The honesty of a diagnosis requires it to answer the counterfactual question: are there cases of successful resistance to pathological recursion?

The historical counterfactual. The "state-delegated, discretionary" organization of science described by Whitley (2010)—roughly Britain from 1950 to 1980—is the case used for comparison here: researchers retained relatively high autonomy from state agencies and local administrative hierarchies; block grants provided comparatively predictable recurrent funding for teaching and research; and scientists could pursue reputations in their specialist fields over the medium term. The source also cautions against exaggerating this autonomy. This type subsequently shifted toward the RAE under political pressures for accountability, selectivity, and concentration in public spending. Martin and Whitley (2010) also document the active roles of university managers and epistemic elites in this transformation and explicitly note that the RAE was not the sole influence. The case therefore supports the claim that external governance pressure changed the buffering conditions, but it does not by itself prove that internal functionality never failed or that every buffer has been removed. The conditions for resisting pathologization must be traced through both the broader political-economic configuration and action within academic organizations.

The technological counterfactual. §1.5 has already unfolded this: Harnad's green self-archiving path and preprint platforms decouple dissemination from evaluation, but about 70% of bioRxiv preprints are still published in traditional journals within two years (Sever et al., 2019), coexisting with rather than replacing them in that sample—the technological means bypassed the commercial publishers' intermediary position without bypassing the deeper discretizing operation of "publication count = academic value."

The two counterfactual groups together reveal: in the cases examined in this chapter, not a single case shows that pathological recursion has been reversed at the system level. This result cannot by itself identify the current state of the suppressing conditions: whether constraints exist, whether they have failed, and why they were insufficient to change the feedback must be traced separately; the removal of buffers is only one explanation compatible with the observation. The diagnosis therefore cannot stop at exposure, nor stop at showing that participants already know the problem—Collyer in §3.3 and the confessions in §3.2 have already proved that "knowing" is not "being able to change." The way out lies not in a cleaner count but in reconnecting judgment to verification—this is precisely the task Part III unfolds in the language of the fiber framework.

§4 Cross-National Comparison: The Same Pressures, Different Institutional Responses

The national distribution of global publication inflation is not even. In 2023, China, the United States, India, and Germany each produced more than 100,000 papers, together over half the world total; between 2014 and 2023 global annual output rose from about 2.2 million to about 3.3 million, with 51% of the increment from China and 13% from India, while output in the United States, the United Kingdom, and Japan declined after their 2021 peak (National Science Board, 2025). The global publication increment was highly concentrated in China and India. This establishes an uneven national distribution of output growth, but a "catch-up logic" must be demonstrated from each country's evaluation and funding arrangements, not inferred from the 51% and 13% shares alone. This chapter traces the Chinese case in detail and does not extend that inference to

India. Catch-up logic also means: the same set of international indicators (citation databases, journal-ranking tiers, world rankings) is plugged into different incentive structures in different institutions, producing similar publication pressures and different forms of distortion. The table compares institutional cross-sections from different historical periods, not a current overview of each country's system.

Country	Main driving force	Institutional response	Key features
United Kingdom	National evaluation (RAE → REF) + commercial publishing	The state actively organizes rankings into a tool of governance	An early case of national audit institutionalization among those examined in this chapter (Jarratt, 1985)
United States	Academic capitalism + tuition model	Marketized restructuring underwritten by public subsidy	Expansion of non-tenure-track faculty; growing managerial power
Italy	The 2010 evaluation reform (ANVUR)	National bibliometric assessment	Collective strategic self-citation; citation clubs
China	Catch-up logic + international index pressure	National excellence initiatives linked to global rankings	Cash rewards per paper; age-based elimination; internalization of indicators
Brazil	WoS coverage bias + global rankings	A national journal-assessment framework (Qualis) correcting coverage bias	Pluralist incentives partially preserved in the economics case

The United Kingdom concentrates the question on how national evaluation changed the behavior of departments and researchers: the Jarratt Report of 1985 reorganized universities as corporate structures, and the RAE evolved from seven-point marking into a mechanism of resource concentration (the degree of concentration by 2001 is given in §2.1)—evaluation shifted from an instrument of community self-governance to an instrument of state governance (Shore & Wright, 2024). The American material requires attending simultaneously to market revenue, public subsidy, and university organization: academic capitalism institutionalized the boundary between university and market, so that prestige accumulation and commercial income reinforce each other within the same organization (Slaughter & Rhoades, 2004). Italy's citation shifts suggest, in turn, that an evaluation reform can change the production of a specific signal—the abnormal rise in Italian inwardness after the ANVUR reform points to collective strategic citation (Baccini et al., 2019; see §2.3).

The Chinese case displays the catch-up logic most completely. In 1998 the Chinese Academy of Sciences launched its Knowledge Innovation Program: in the first stage, the 76 targeted research institutes cut their staff by 50%, with the worst performers—especially those over fifty—eliminated; SCI-paper bonuses ranged from $240 to $1,200, Beijing Astronomical Observatory's SCI papers rose from 25 in 1998 to more than 70 in 2000, and its graduate students had to publish four SCI papers to graduate (Ding, 2001). A qualitative study published by Tian et al. in 2016 found that the pressure had been internalized from institutional policy into everyday experience: interviews with seven young natural-science scholars at a research university showed

that the requirement to publish in internationally indexed journals made them rush papers out with no time for training in teaching, that writing time squeezed out family life, and that the interviewees themselves reported finding it harder to produce high-quality or genuinely novel results (Tian et al., 2016). In February 2020, the Ministry of Education and the Ministry of Science and Technology issued a directive on standardizing the use of SCI-paper indicators in higher education, requiring the cancellation of rewards granted directly on the basis of such indicators, prohibiting their direct linkage with resource allocation, and disallowing publication counts and impact factors as restrictive conditions for degree conferral (Ministry of Education, & Ministry of Science and Technology, 2020)—whether the policy changed actual assessment requires separate examination, but it at least shows that the Chinese case cannot be written as a one-way deepening of indicator pressure. By 2023, China accounted for 51% of the global publication increment (National Science Board, 2025)—the three bodies of material display institutional rewards, researchers' lived experience, and national output growth respectively; together they indicate the importance of evaluative incentives, but their statistical units and time frames differ and cannot be pieced into a single continuous curve.

Among the cases examined in this chapter, Brazil offers a partial example of preserved evaluative pluralism. Facing the missing journal coverage and low counts of the Web of Science for the humanities and social sciences (Almeida et al., 2018; see §3.5), Brazil used the national journal-assessment framework Qualis to correct the coverage bias of the international indexes; analysis of the 2016 revision of Qualis shows that this correction preserved pluralistic incentives in economics—journals of heterodox traditions kept their places in the higher strata (Almeida et al., 2018). The conditions for protecting pluralism this chapter summarizes in three points: the evaluator is locally and continuously maintained; heterodox scholars participate in setting the stratification standards; and the link to international rankings is loosened but not severed. The contrast is thereby clear: in the United Kingdom and Italy the evaluator is supplied by international commercial indexes or national assessment, while Brazil retained local evaluative authority; yet what an evaluator protects is decided not only by who maintains it and to whom it answers, but also by which scholarly communities take part in setting standards, which contributions gain recognition, and whether dissent can change stratification outcomes.

The cross-national cases do not share a single national culture or institutional starting point; when external indicators acquire exclusive authority and displace substantive verification, these cases exhibit stratification and distortion of different forms and intensities. They do not establish that any use of external indicators necessarily produces distortion; the decisive questions are how verification conditions change and whether counter-evidence and objections can alter subsequent judgments. Institutional responses to the same global pressures do differ across countries; such differences change the binding force of indicators, researchers' modes of adaptation, and the space for independent judgment—their role is not limited to altering the speed and intensity of pathologization. Zhao Dingxin's critique of academic institutional mimicry applies here: establishing a set of internationally familiar publication requirements does not mean the corresponding conditions of research criticism and autonomy have also formed; conversely, a local system cannot, merely by differing from international rankings, prove that it has protected the diversity of knowledge (Zhao, 2004, 2022). The use of cross-national comparison is to find which relations are variable and which conditions make intervention effective—this chapter does not rank countries by academic quality on the basis of limited cases, nor does it declare any single system the only successful resistance.

Part I ends here. These mechanisms may be constrained by transnational publishing, ranking, and professional networks and can therefore be difficult to reverse within a single national framework; this chapter does not

make the universal claim that they are difficult to reverse in every individual country. But these diagnoses are scattered across different disciplines, terminologies, and levels of analysis—historians of the industry speak of the "unit of knowledge" becoming an "accounting unit," econometricians of "Goodhart's law," sociologists of "reactivity," epistemologists of the "hermeneutic circle." They describe one and the same structural problem, yet lack a unified analytical language to name and locate its core mechanism. This is precisely the motivation for introducing the fiber framework in Part II.

Part II: A Review of the Fiber Framework

§5 Core Concepts

This section reviews the ten entries of the fiber framework: five independently load-bearing core concepts (fiber, pseudo-small fiber, dynamic corrosion, cognitive debt, narrative load) and five auxiliary entries carrying mathematical representation, mechanism explanation, the base conceptual layer, and self-examination—the deletion test eliminates only entries that neither bear load nor carry a function; the self-referential audit (box below) acknowledges that "ten" bears no load, and the ten-item presentation is kept for the trackability of the argument. Most of the concepts already have strict definitions in Chapters 1 through 4 of this book; this section reviews them by definitional restatement plus intuitive elaboration, assuming the reader will not flip back to the chapters cited.

> **Self-referential audit: the impulse to count.** This chapter criticizes the "salami slicing" strategy described by Bornmann—cutting research into countable thin slices to fit the indicator—yet is the precisely countable list of "ten items" not itself an instance of the same impulse to count? The honest answer is: yes. On self-examination, the truly independent load-bearing core among the ten is five (fiber, pseudo-small fiber, dynamic corrosion, cognitive debt, narrative load); the remainder are mergeable mechanism explanations (visible/hidden subspace → pseudo-small fiber), information-theoretic bridges (ontological dimension reduction—its information-theoretic formulation is Chapter 4 §1, Proposition 3), structural twins (mapping function ← fiber), a base conceptual layer (time/clock dimension → discretization), and a self-referential tool (the baroque). The ten-item presentation is kept for the trackability of the argument, but the reader should be aware: the number "ten" bears no load—it may itself be an instance of the behavioral pattern the framework diagnoses. The hierarchical relations of the concepts are marked in the sections that follow.

§5.1 The Fiber

Definition. Given a mapping function $\pi : \mathcal{T} \to \mathcal{O}$, the fiber of a reading $r \in \mathcal{O}$ is $\pi^{-1}(r) = \{T' \in \mathcal{T} \mid \pi(T') = r\}$—the set of all true states that could have produced that reading.

The fiber is a subset of the space of true states. The larger the fiber, the less information the reading provides; a singleton fiber means the reading uniquely determines the truth; a fiber equal to the whole space means the reading is compatible with every possibility and carries zero information. The fiber is an objective mathematical object—its size does not depend on the observer's psychological state, only on the structure of the mapping function. The mapping function itself, however, may be partly unknown, in which case what we discuss is the set of true states compatible with the observations under the known constraints—the "identification region" in Manski's (2003) partial identification framework: when observational evidence cannot uniquely determine the

true value of a parameter, the set within which the parameter is confined; withdrawing from the pretense of point identification and the honest labeling of uncertainty boundaries is precisely the precursor of the "annotating the fiber" path in statistical inference. Failure to recover every feature of an object does not imply failure to answer a specified question: a reading is sufficient for that question when all compatible states give the same answer (the boundary condition of Chapter 1 §4.3 and Chapter 4 §1, Proposition 3).

§5.2 The Mapping Function

Definition. The mapping function (π) encodes the entire correspondence from true states to readings—the precision of instruments, the limits of perception, the way data are processed, the filtering effects of theoretical presuppositions.

The mapping function is the structural twin of the fiber: the fiber is the inverse image for a given reading, $\pi^{-1}(r)$; the mapping function is π. Without the mapping function there would be no concept of a fiber. The key diagnostic variable is not the number of readings but the degree to which the mapping function is known.

> **Hierarchy note:** the mapping function and the fiber define each other (π and π^{-1}) and in the strict sense could be merged into the definition of the fiber in §5.1. It is listed separately because the diagnostic variable "degree to which the mapping function is known" is more operational in practice than fiber size—we can judge that "the mapping function has been rewritten by its use" (dynamic corrosion), whereas the absolute size of a fiber is usually not directly measurable (see the methodological note in §5.6).

§5.3 Dynamic Corrosion

Proposition. When readings are used for reward, punishment, resource allocation, or performance assessment and actors can adjust around them, the empirical relation between readings and the evaluative target may change through feedback from their use; this book calls such relational drift dynamic corrosion. Strictly speaking, what changes is the empirical relation between readings and the evaluative target: the computational rule can remain unchanged while the composition of output and the correspondence have already changed; "the mapping function rewritten by its use" is shorthand for this process (the original formulation in Chapter 1 §4.2 is "the mapping function can no longer be treated as a permanently stable exogenous parameter").

This proposition is not contained in the definition of the fiber. The definition of the fiber is a set-theoretic proposition; dynamic corrosion is a temporal-causal proposition. Goodhart, Campbell, Espeland and Sauder, Scott, and Zhao Dingxin (Goodhart, 1975/1984; Campbell, 1979; Espeland & Sauder, 2007; Scott, 1998; Zhao, 2022) supply comparable mechanism materials from different domains, but these materials cannot be counted as five independent confirmations: Espeland and Sauder explicitly connect reactivity with Campbell's discussion of reactive measurement. Together they support a conditional temporal-causal mechanism, not a universal law that every indicator must be gamed. Dynamic corrosion comprises two mechanisms of drift: strategic gaming (the intentional manipulation of readings) and selective adaptation (actors who do not adapt to the indicator are progressively eliminated). The latter is harder to defend against—because there is no identifiable agent to punish.

§5.4 Ontological Dimension Reduction

Proposition. If the dimension of the space of true states $\mathcal{T}$ exceeds that of the observable space $\mathcal{O}$ ($\dim(\mathcal{T}) > \dim(\mathcal{O})$), then mappings of the kind used for evaluation (under the usual topological assumptions and continuity conditions) are necessarily non-injective—there exist distinct true states that share the same reading, and hence non-singleton fibers; which readings have non-singleton fibers depends on how the mapping discards dimensions.

The data-processing inequality of information theory ($I(X;Z) \leq I(X;Y)$ for a Markov chain $X \to Y \to Z$) gives the information-theoretic formulation of this limit (equality is permitted; see Chapter 4 §1, Proposition 3): information obtained after processing through intermediate links cannot exceed the original information; it shows that the loss of a lossy mapping cannot be recovered by post-processing, without by itself proving that every reduction is strictly lossy. Even knowing every detail of the mapping function perfectly, no inverse mapping exists that uniquely recovers all true states (Chapter 1 §4.3)—the fiber of an individual reading may still be a singleton.

> **Hierarchy note:** ontological dimension reduction is the information-theoretic counterpart of the dimensional limit (equality permitted; see Chapter 4 §1, Proposition 3), not an independently load-bearing concept. The dimensional argument (when $\dim(\mathcal{T}) > \dim(\mathcal{O})$, continuous mappings cannot be injective, so non-singleton fibers exist) already entails that "high-dimensional truth cannot be globally and uniquely determined by low-dimensional readings"; the data-processing inequality supplies information-theoretic support for this direction (equality allowed; see the formulation in Chapter 4 §1, Proposition 3). It is listed separately to bridge the definition of the fiber (set theory) and the information-theoretic tradition, so that readers of both disciplines can find their entry.

§5.5 Visible and Hidden Subspaces

Definition. Relative to the state of knowledge of a designated reference observer, the space of true states can be conceptually partitioned into a visible subspace ($\mathcal{T}_v$, where the observer has relatively adequate knowledge and fibers can be constrained small enough within this subspace) and a hidden subspace ($\mathcal{T}_h$, where the observer lacks effective discriminating power and fibers within this subspace may be large).

This decomposition is the mechanism explanation of the pseudo-small fiber: when the system's optimization objective is to maximize some preference signal on the visible subspace, the system learns to fit the target distribution on the visible subspace; but it lacks effective constraint on the hidden subspace—changes in the hidden dimensions receive no constraint from that optimization objective. One key corollary is: **repairing visible problems may raise the relative weight of hidden bias.** "Using a clock to calibrate a clock is not calibration" is a direct corollary of this decomposition: if all mappings share the same systematic bias, cross-validation on the visible subspace yields only internal consistency, not alignment with the truth. Consistency is not calibration.

> **Hierarchy note:** the visible/hidden subspaces are the mechanism explanation of the pseudo-small fiber (§5.6) and in the strict sense could be merged into §5.6. They are listed separately because the corollary "repairing visible problems may raise the relative weight of hidden bias" is extremely important in practice—it explains why the "improvements" of indicator governance cannot by themselves prove overall reliability—whether the absolute level of hidden bias has changed requires

separate verification; only when visible improvements breed overconfidence or crowd out other checks can the system become more fragile even as it improves locally.

§5.6 The Pseudo-Small Fiber

Definition. Let the system's true fiber be $F_{\text{true}} = \pi^{-1}(r)$, and the perceived fiber formed by the observer on the basis of visible-subspace information be $F_{\text{perceived}}$. When F_{true} is large but $F_{\text{perceived}}$ is small, the fiber is called a pseudo-small fiber.

The pseudo-small fiber must be clearly distinguished from two other states: the truly small fiber (F_{true} and $F_{\text{perceived}}$ both small and aligned) and the truly large fiber (F_{true} and $F_{\text{perceived}}$ both large, the system candidly acknowledging uncertainty). The core feature of the pseudo-small fiber is a systematic disconnect between an appearance of certainty and a substance of uncertainty. The first two states, even when their conclusions are wrong, preserve the possibility of audit; the particular danger of the pseudo-small fiber is that its appearance of certainty reduces the observer's incentive to seek further checks—without eliminating the possibility that external audit identifies it.

A pseudo-small fiber cannot be directly detected from the content of the system's output alone—for the misled observer, the appearance of certainty does not reveal the size of the true fiber; and an analyst's recognition that the true fiber is large does not by itself undo the misled recipient's earlier perception of certainty: being misled and being diagnosed are different positions at different moments. But pseudo-small fibers can be inferred structurally: once the analyst identifies the structural conditions under which pseudo-small fibers most readily form (the optimization objective driven only by preference signals, the absence of independent calibration sources, the observer passively receiving readings), there is reason to launch a precautionary audit of these conditions, even if no particular instance can be pointed to—a preventive decision, not the finding that pseudo-small fibers are already being produced.

> **Methodological note (ordinal comparison)**
>
> The absolute size of a fiber usually cannot be measured precisely—that would require complete knowledge of the mapping function and of the structure of the space of true states. But the real value of the fiber framework lies in ordinal comparison: on the basis of partial knowledge of the mapping function, making grounded judgments—whether fiber A is larger than fiber B. Ordinal comparison is a practical method rather than a load-bearing concept; it suffices to support most practical judgments, and is therefore demoted to a methodological note without taking a concept number; where containment or a common measure is absent, the comparison should either state its basis or be marked as not yet comparable, rather than forced into an ordering (Chapter 4 §2.3).

§5.7 Narrative Load

Definition. Narrative load (ϕ) is the narrative packaging and information asymmetry projected by the projecting party in pursuit of a local optimum. Its function is to alter the recipient's perception of the reading—to make a large fiber look like a small one.

The temporal behavior of narrative load has two orders. Its first-order change corresponds to observable narrative switching—visible, auditable. Its second-order change (borrowing the metaphor of calculus: the accelerated consolidation of underlying narrative frames, Chapter 3 §2.3) corresponds to the formation of industry

defaults and even civilizational defaults—by uniformly covering the information environment, it abolishes the contrast conditions that audit requires. Narrative load is the bridge between information structure and social action. Without it, the framework could diagnose "the fiber is large" and "the system looks certain," but could not explain "why the system invests in making large fibers look small"—who is driving this?

§5.8 Cognitive Debt and Shadow Debt

When pseudo-small fibers are systematically produced and accumulate over time, the system enters a state of "cognitive debt": the true fiber is expanding or holding steady, but the observable signals show that all is well. The gap between true cognitive debt (D_c^{true}) and observable cognitive debt (D_c^{obs}) is the shadow debt ($D_c^{shadow} = D_c^{true} - D_c^{obs}$)—it cannot be directly observed, but when it accumulates to a critical point, a "liquidation" occurs. The tracking of shadow debt gives the framework its temporal dimension.

Cognitive debt follows the "accumulation → concealment → liquidation" three-stage lifecycle. **Accumulation stage:** the high narrative load of the projection side obscures the visibility of the liability, and each positive increment of cognitive debt is deposited in interaction. **Concealment stage:** intertemporal feedback gives the accumulated debt a "compound-interest" character—cognitive debt not liquidated in the current period keeps eroding the effective capacity of the defense side, reducing next-period defensive capacity and accelerating new debt. **Liquidation stage:** when accumulated debt breaks through the stress limit of the legitimacy base, or some exogenous shock forces the implicit liability to be honored—the debt clock matures, and implicit debt converts from hidden book liability into explicit loss. Liquidation is not the disappearance of debt; it is the forced conversion of the liability from implicit to explicit form.

This lifecycle is governed by a covering law that Zhao Dingxin calls "the law of strength-weakness transformation" (the covering law behind "Daoist time" in Chapter 3): "any social organization, idea, or institution, as it grows strong, will face increasingly important social forces and mechanisms that weaken it" (Zhao, 2019; author's translation). The exponential expansion of the projection side accumulates cognitive debt through the concealment stage, but as the projection side grows strong, the mechanisms that weaken it grow ever more important. In the language of this book's framework, these weakening mechanisms appear as follows: once the first derivative of narrative load becomes visible, adaptive offsets are triggered; the buffer of the legitimacy base is consumed in ways that mask the decay of capability. This law constitutes the structural pressure under which the concealment stage can hardly last unconditionally—but pressure converts into liquidation only under specific conditions: weakening forces may form a stable constraint without triggering concentrated liquidation, and losses may long remain dispersed; liquidation is not an accidental event, yet its scope and form depend on institutional conditions.

§5.9 The Time Dimension and the Clock Dimension

The **time dimension** is a continuous, multichannel, relational mode of being. The body simultaneously receives signals on independent channels—temperature, light, hunger, fatigue—and these channels can cross-check one another. The **clock dimension** is a discrete, scaled, reading-only mode of operation. One number, no corroboration. Moving from the time dimension into the clock dimension is not a neutral switching of operating modes—the three limitations stack here and may reinforce one another: single-channel operation without cross-checking means independent calibration no longer supplies new constraints (when the reading and existing constraints exclude no candidate state, the compatible set is the whole space; Chapter 1 §5); taking readings

as the basis of decisions means the mapping function is rewritten by use (dynamic corrosion); compressing continuous multidimensional truth into discrete finite-dimensional readings may discard distinctions relevant to the judgment at issue; whether the compression is strictly lossy must be examined against the state support, the mapping, and the question being asked (ontological dimension reduction). The three layers are not three parallel problems but one self-enclosing trap.

Chapter 3 defines cognitive friction ($\mu_c = A \cdot E_c$) as a necessary condition of maintaining subjectivity—its absence has two paths, the automation paradox and algorithmic *Xiangyuan*; the work of meta-audit manifests operationally as contacts that cannot be dispensed with—attention, pause, and discernment. Its point of application in this chapter is not retaining red tape but retaining the contacts that can change the direction of research: redoing an analysis after reading a counterexample, suspending a conclusion in the face of a gap in the material, allocating time to a re-examination that yields no new paper. Automation that reduces formatting labor while preserving these contacts can strengthen judgment; automation that hands over to the old readings even the question whether judgment needs checking erodes the working conditions of meta-audit—the closing question of §9.3, "who retains contact with the original question, and who holds the right to stop the process and re-verify," is precisely the unfolding of this distinction in the AI scenario.

> **Hierarchy note:** the time/clock dimension is the base conceptual layer of the discussion of discretization. §0.2 gives the definition of discretization from morphology (four ways of cutting); this section gives the mechanism from mode-switching (continuous multichannel → discrete single channel); Part III will give the operationalization in the concrete "publish or perish" scenario—the three are corresponding formulations of one operation at three levels of abstraction.

§5.10 The Baroque

Von Neumann (1947) warned in "The Mathematician": a discipline that moves far from its empirical sources and undergoes much "abstract" inbreeding is in danger of degeneration—"At the inception the style is usually classical; when it shows signs of becoming baroque, then the danger signal is up." The concept of the baroque is the framework's immune system—it lets the framework ask of itself: "Is my proliferation of concepts multiplying far from the empirical objects?"

> **Hierarchy note:** the baroque provides no independent diagnostic capacity with respect to "publish or perish" or any external object—it is a self-referential tool, not a diagnostic tool. It is retained because the self-referential examination of Part IV (§11, auditing this chapter itself as a mapping operation) needs it as an operational lever ("is the deletion test itself also a mapping operation?"). But the baroque bears no load—removing it would cost the framework none of its diagnostic power over the publication system, only its diagnostic power over itself. The self-examination of the impulse to count at the opening of §5 is an instance of the baroque concept applied to the framework itself.

§6 Academic Lineage

The mathematical object used in this chapter is the set-theoretic inverse image $\pi^{-1}(r)$, which does not rely on the local trivializations or structure groups of Steenrod's (1951) fiber bundles; Steenrod marks the mathematical lineage of this notation. Over recent decades, the concept has undergone independent development in several

disciplines.

Measurement theory. Representational measurement theory (Krantz et al., 1971) holds that measurement is a homomorphic process of mapping an empirical relational system into a numerical relational system—numerical assignment need not be one-to-one: two rods of equal length are different empirical objects yet receive the same number. The reliability of measurement depends not only on instrumental precision but also on the calibration relation between the mapping function and the truth—precisely the theoretical precursor of the fiber framework's distinction between "fiber size" and "pseudo-small fiber."

Econometrics. Manski's (2003) method of "partial identification" demonstrates that when the distribution of observed variables cannot uniquely determine the true value of a parameter, the parameter is confined to a set compatible with the observations—the identification region—rather than to a single point. Manski's withdrawal from the pretense of point identification and his honest labeling of uncertainty boundaries are the exact precursor of the "annotating the fiber" path in statistical inference.

Information theory. Shannon's (1948) conditional entropy $H(X|Y = y)$ measures the probabilistic uncertainty still remaining after a given observation—when the state space carries a probability distribution, it can serve as one measure of fiber size (one of the three measures in Chapter 4 §2.1). The data-processing inequality gives the information-theoretic formulation of the §5.4 proposition (equality permitted; see Chapter 4 §1, Proposition 3). The information-bottleneck method of Tishby et al. (1999) further formulates the problem as a trade-off between compact representation and the preservation of relevant information.

Machine learning. Conformal prediction, developed by Vovk et al. (2005), provides the most direct engineering response to the problem of "annotating the fiber"—under weak conditions such as exchangeability, without presupposing any specific form of the data distribution, it generates prediction sets that contain the true label with a specified probability—generally a marginal guarantee of average coverage, not an identical conditional probability for every case; the width of the prediction set can serve as an engineering reading of the magnitude of uncertainty. By contrast, asserting an unidentified quantity as identified manufactures pseudo-small fibers; conformal prediction makes uncertainty explicit under valid coverage guarantees—though set-valued form does not by itself provide that honesty.

Sociology. Zhao Dingxin's (2022) theory of the "hermeneutic circle" describes the epistemological loop between local information and holistic understanding. The "illusion of closure" of the hermeneutic circle corresponds exactly to the pseudo-small fiber of the fiber framework. Their diagnoses agree: consistency is not calibration, and confidence is not correctness.

The academic positioning of this book's fiber framework does not lie in having invented "the fiber as an epistemological tool"—that insight has been discovered independently several times. The book's contribution lies in constructing a unified analytical tool with the "pseudo-small fiber" as its core diagnostic concept, oriented toward uncertainty management and cognitive-debt accountability, and in explicitly directing it toward cross-domain application.

Part III: Diagnosis in the Fiber Framework

§7 The Fiber Diagnosis of the Academic Publication Indicator System

The history and diagnosis organized in Part I can be reread in the unified language of the fiber framework. "Publish or perish" is not only a complete mapping system—the quality and contribution of academic research (high-dimensional true state) replaced by quantitative indicators such as publication counts, citation counts, and impact factors (low-dimensional readings)—it also satisfies the combination of conditions the fiber framework requires for diagnosis: high-dimensional truth reduced to low-dimensional readings, readings wired into resource allocation, the absence of an independent calibration channel, and evaluation pressure continuously reshaping output.

§7.1 "Publish or Perish" as a Mapping System

The state considered here includes the research and its publication, dissemination, and use; quality and contribution are the aspects to be assessed. The mapping function converts selected records into publication counts, citation counts, or other readings—the question is which quality-relevant distinctions those readings preserve and which they merge. The readings are the ordering parameters that personnel systems, funding agencies, and ranking agencies can use. This chapter models academic quality and contribution as a state space containing multiple distinctions to be preserved. If the evaluative task requires retaining those dimensions and the mapping conditions of §5.4 hold, a single number cannot globally determine quality uniquely. If only a specified question is asked, §5.1 requires checking whether all compatible states give the same answer.

The history organized in Part I can now be precisely located: professional reputation already affected appointments and resources in earlier academic institutions; the formation of the functional circuit in §1.3 corresponds to the stage in which quantitative indicators had not yet acquired exclusive allocative authority. Later reforms changed the weight, scope, and binding force of indicators—the question to track is not when resource feedback first appeared, but which metrics gained greater allocative power and whether substantive scrutiny strengthened alongside them. The pathologizing mechanisms analyzed in §§1.4–1.5 correspond to the stage after the mapping function entered decision-making and dynamic corrosion began. The binding of the three functional lines (knowledge circulation + prestige accumulation + commercial profit) in §1.4 is precisely the institutional carrier of "the mapping function rewritten by its own use."

§7.2 How Misuse of the Journal Impact Factor Creates Pseudo-Small Fibers

A pseudo-small fiber can arise when a journal-level impact factor is treated as proof of an individual paper's reliability.

The true fiber is large. The relation between a journal's impact factor and the quality of the research it publishes is highly many-to-many. High-impact journals may publish much eye-catching but irreproducible research; low-impact journals may publish slow-burning foundational work that eventually remakes the field. McGilchrist, reviewing Munafò and Flint's tracking of "breakthrough" research in medical genetics, notes that studies published in high-impact journals such as *Nature* and *Science* were in fact less likely to give accurate effect-size estimates than studies in lower-impact journals—in that field, there is even a negative correlation between impact factor and the reliability of effect-size estimates (McGilchrist, 2021). Kamat and Schatz point out that

"nearly 60% of the papers published in any given journal are cited less than the average citation per paper published during the two-year JIF period" (Kamat & Schatz, 2015)—a journal's impact factor cannot represent the actual influence of most of the papers within it.

The perceived fiber is small. The impact factor long presented itself as a decimal precise to three places (for example, "29.274")—the JCR has used one decimal place since the 2023 edition, but reducing the decimal places has not changed the indicator's level: it remains a journal-level citation statistic from which the reliability of an individual paper cannot be judged. Personnel systems, funding agencies, and ranking agencies make resource-allocation decisions on its basis. When numerical precision is mistaken for sufficiency in quality judgment, the reading generates a sense of certainty beyond its evidential range.

Systematic disconnect. The impact factor is not an individual scholar's cognitive bias; it is structurally produced by the optimization process of the evaluation system: when resource allocation takes the impact factor as its standard, journals have a structural incentive to manipulate it, and scholars have a structural incentive to choose research directions that can be placed in high-impact journals—these behaviors can raise the readings without a corresponding improvement in quality; whether they also change research quality requires separate judgment. Gingras states this precisely as the scientific paper's transformation from a "unit of knowledge" into an "accounting unit" (Gingras, 2020).

§7.3 Goodhart's Law as Dynamic Corrosion

The five different research traditions organized in §5.3—Goodhart, Campbell, Espeland and Sauder, Scott, Zhao Dingxin—share one precise structural diagnosis in the fiber framework: they describe one and the same mechanism—dynamic corrosion.

Erren et al., using game theory, prove that the "publish or perish" scenario takes the form of a prisoner's dilemma under fairly weak assumptions: even though both parties would be better off delaying publication of more complete results, rational researchers will still choose to publish incomplete results early—because the priority payoff is too high (Erren et al., 2015). Agrawal's correlation of 0.85 is compatible with dynamic corrosion, but the preprint itself states that correlation does not imply causality; active investigation, follow-up scrutiny, the retraction-date convention, and the extrapolated 2024 publication volume can all shape the indicator. It is therefore a signal for tracing evaluation incentives, output growth, discovery, and handling capacity, not independent identification of dynamic corrosion. Baccini et al.'s "inwardness" indicator quantifies the collective behavioral expression of dynamic corrosion at the national level (Baccini et al., 2019).

The two mechanisms of dynamic corrosion are simultaneously present in "publish or perish." Strategic gaming: citation cartels, salami slicing, gift authorship without substantive contribution. Selective adaptation: scholars who do not devote their energy to indicator optimization occupy disadvantaged positions in resource allocation—no agent needs to consciously manipulate the indicators for the system to corrode.

§7.4 Rankings as Closed-Loop Pseudo-Calibration

The global university ranking system is an institutional instance of "using a clock to calibrate a clock." Ranking agencies construct comprehensive rankings from indicators such as journal impact factors and citation counts; university managements allocate resources on the basis of rankings; scholars adjust research directions toward rankings; the adjusted academic output then becomes the input of the next round of rankings. Multiple mapping

systems calibrate one another—each system's readings take the other systems' readings as their reference—and the true state (academic quality) partially disappears from the input of the mappings.

Espeland and Sauder, taking law-school rankings as their case, propose the concept of *reactivity*: rankings are reactive measures—as the paper's title says, public measures "recreate social worlds"—they modify the phenomena they measure and change how people understand their situations (Espeland & Sauder, 2007). In fiber language: the ranking reward function constrains readings on the visible subspace; whether activities lacking commensurate channels of scrutiny—teaching, interdisciplinary collaboration, public dissemination—have actually deteriorated requires process evidence. As visible bias is suppressed, the relative weight of hidden bias may rise—that is the applicability condition of the §5.5 corollary, not a measured state.

§7.5 The Institutional Operation of Narrative Load

Within the "publish or perish" system, the operating mechanisms of narrative load can be concretely tracked:

- **Journal brand narratives.** "Published in *Nature* or *Science*" carries a legitimacy far exceeding what the content's quality could support—the brand narrative makes a large fiber (this research may be irreproducible) look like a small fiber ("it was published in *Nature*, therefore it is right"). §3.6 has given McGilchrist's observation—of the intellectual giants he lists, nearly all (excepting Einstein) transmitted their influence through lectures or books rather than papers; when "publish or perish" strips books of "career value," the academic system systematically squeezes out one important medium that carries synthetic argument.
- **Methodological narratives.** Zhao Dingxin's diagnosis of the ills of the interpretive tradition gains new precision in an indicator environment: when evaluation rewards only publishable conceptual innovation, applying fashionable theoretical frames and masking the poverty of content with rhetorical splendor becomes the academic strategy with the highest marginal payoff—grand terminology receives the evaluation of "theoretical depth" on the visible subspace, while explanatory power and testability degrade in the hidden subspace (Zhao, 2004).
- **The narrative of the indicators themselves.** "What is quantifiable is scientific"—this narrative is the deepest layer of narrative load; its historical formation was given in the audit-culture lineage of §2.1. It grants legitimacy to the pseudo-small-fiber state of the entire evaluation system: not "we are replacing academic judgment with a lossy mapping," but "we are replacing subjective bias with objective indicators."

§7.6 The Three Clocks and the Dynamics of Cognitive Debt

This chapter uses the techno-capital clock (τ_c), the governance clock (τ_g), and the debt clock (τ_d) to distinguish three processes: expansion, institutional response, and the first manifestation of problems. Their relative time scales are relations to be diagnosed in the dynamics of cognitive debt, not a fixed ordering shared by all academic settings.

The techno-capital clock (τ_c) is the characteristic period of the expansion of the projection side of "publish or perish"—the time scale over which publication pressure, indicator proliferation, and commercial expansion complete one round of mutual amplification. Competitive selection pressure may shorten some production and adoption cycles, but the dates on which the h-index and the impact factor were introduced and diffused are

milestones of institutional adoption, not measurements of the techno-capital clock's period. Whether successive rounds accelerate requires separate longitudinal process evidence. The "accumulative development time" of Zhao Dingxin (2019) (§3.5) gives this clock its sociological character: directional, accumulative, generating isomorphic pressures.

The governance clock (τ_g) is the full response cycle of the institutional level to the pathology of "publish or perish"—from identifying the problem, verifying evidence, forming a plan, to implementing an intervention. Governance responses require time for evidence formation, procedural coordination, and implementation. Zhao's diversity time can characterize aspects of their institutional environment, but it does not yield a uniform incompressible lower bound. Response cycles can differ substantially across interventions; REF/TEF (Research/Teaching Excellence Framework), open access, and DORA (the San Francisco Declaration on Research Assessment) cannot all be declared far slower than metric proliferation without comparison.

The debt clock (τ_d) is the characteristic delay of cognitive debt from sedimentation to manifestation. A study that is insufficiently verified deposits a verification gap at the moment of publication; the gap surfaces through replication attempts, citation accumulation, and even the consequences of application—the delay can vary substantially with the research task, the correction channel, and the triggering event; some cases may extend over decades or generations, while other errors surface in shorter cycles of replication, correction, or retraction. The interpretation of a failed replication must still distinguish an erroneous conclusion, different conditions of applicability, and limitations of the replication itself. The debt clock is hard to predict until the trigger condition appears; what collapses to zero at that moment is the remaining waiting time, not the elapsed deposition period—systemic crises (such as the reproducibility crisis) are a forced, concentrated realization of previously latent liabilities. Warning signs such as shrinking verification windows and deteriorating proxy indicators may appear in advance, but the trigger moment cannot be precisely predicted.

Structural ordering and temporal explanation. In what this chapter calls the normal scenario, if the cycle of techno-capital expansion is shorter than the governance response, and the first manifestation of debt occurs later still, the ordering $\tau_c \ll \tau_g \ll \tau_d$ may hold. This ordering is a relation to be diagnosed, not a presumed universal frequency. The diffusion of a metric, behavioral adaptation, the accumulation of problems, and their detection are distinct processes; their start and end events must be specified and their characteristic time scales traced separately. The ordering cannot be obtained by subtracting the year an indicator was created from the year of some critical paper, nor by importing fixed scales of months, years, or generations. The relative positions of the governance and debt clocks require case-specific judgment: local correction may precede institutional response, and institutional intervention may precede concentrated losses. Only when the manifestation of debt precedes the completion of the governance response does the crisis situation described by Collingridge's (1980) dilemma arise—governance then faces losses already exposed.

The accumulation → concealment → liquidation mapping of "publish or perish." The three-stage lifecycle established in §5.8 has an analytical comparison in "publish or perish." **Accumulation stage:** in the early years after the impact factor's introduction, publication volume grows while quality erosion is not yet visible—the techno-capital clock is running, the governance clock not yet responding; this stage description requires historical evidence and is not a fact automatically supplied by the three-stage model. **Concealment stage:** intertemporal feedback makes accumulated debt grow at "compound interest"—irreproducible studies remain in the literature, continuing to be cited and applied, while new irreproducible studies continue to be produced under the legitimizing cover of the previous batch. Meta-audit capacity is systemically dismantled

at this stage: replication studies are not seen as "original contributions," and scholars' motivation to verify is dissolved by the incentive structure. **Liquidation stage:** the high irreproducibility rates summarized by Begley and Ioannidis (75–80% of 67 studies; about 90% of 53 oncology studies), Park et al.'s decline of disruptiveness, Agrawal's correlation of 0.85—these studies identify different problems in research reliability, citation structure, and retraction records. The common explanation proposed here is that some evaluation incentives expand output without sustaining verification; establishing a shared pathological loop requires tracing the links among assessment, production, use, and correction. Liquidation is not the disappearance of debt—retracted papers continue to be cited (Agarwal & Ioannidis, 2019); implicit liability is exposed in the form of explicit loss, but the underlying incentive structure is unchanged, and a new batch of debt is already accumulating. The law of strength-weakness transformation already cited in §5.8 constitutes the structural pressure of this rhythm: the stronger the projection side, the more important the mechanisms that weaken it—the growing exposure surface triggers adaptive gaming, and the legitimacy buffer is consumed; yet whether weakening forces convert into concentrated liquidation still depends on concrete conditions, and cannot be inferred merely from the diminishing marginal effect after noise saturation. The concealment stage can hardly last unconditionally, but after liquidation the system does not automatically enter a "zero-liability" state—it enters a new round of accumulation.

§7.7 Discretization as the Meta-Operation of "Publish or Perish"

The one structural action running through the diagnosis above is **discretization**—of the four forms of discretization defined in §0.2 (commensuration, unitization, time-slicing, API-fication), what operates in the "publish or perish" scenario is **commensuration**. The quality of academic research (the continuous multidimensional true state—originality, methodological rigor, logical force of argument, long-term influence on the field, contributions to teaching) is discretized into countable, rankable, comparable quantitative indicators (paper counts, citation counts, impact factors, the h-index). Discretization itself is not the lesion—compressing multidimensional quality into finite-dimensional readings is an operation no evaluation system can avoid. The lesion is the usurpation of the instrumental-rational result of discretization as the value-rational essence—taking "publication count" as "academic value" itself, taking "impact factor" as "research quality" itself—that is the epistemological root of pathological recursion.

§5.9 established the distinction between the time dimension and the clock dimension: the time dimension is a continuous, multichannel, cross-checkable mode; the clock dimension is a discrete single-channel mode without corroboration. The discretizing operation of "publish or perish" forcibly transfers academic evaluation from the time dimension to the clock dimension: one impact-factor number replaces the multidimensional judgment of research quality; one h-index replaces the peer assessment of a scholar's long-term contribution. The three limitations stack here and may reinforce one another: single-channel operation without cross-checking (when the reading and existing constraints exclude no candidate state, the compatible set is the whole space, Chapter 1 §5), readings taken as the basis of decisions (dynamic corrosion), continuous multidimensional truth compressed into discrete finite-dimensional readings (ontological dimension reduction).

Gingras states this precisely as the scientific paper's transformation from "unit of knowledge" into "accounting unit" (Gingras, 2020)—the unit of knowledge is multidimensional, qualitative, embedded in context; the accounting unit is one-dimensional, quantitative, detached from context. That the discretizing operation lets the latter replace the former is not technical progress but dimension reduction at the ontological level.

The diagnostic position of preprint platforms (arXiv, bioRxiv, medRxiv) in the fiber framework deserves separate discussion. Preprints decouple dissemination from evaluation—researchers can share work without waiting for peer review and journal certification (Harnad, 1990, 2008; Sever et al., 2019)—which formally appears to be a resistance to the journal monopoly. But stated precisely in the framework's language: preprints lower the barrier to accessing material and to raising objections; career advancement still takes journal publication as its standard, but whether the earlier scrutiny enters subsequent judgments determines the preprint's actual effect on evaluation. About 70% of bioRxiv preprints are published in traditional journals within two years of first posting (Sever et al., 2019; see §1.5)—in that sample they coexist with journals rather than replace them, a parallel channel in dissemination whose certification effect depends on those conditions. More complex still, preprints raise the density of information arrival—scholars must track not only formally published papers but an ever-growing stream of preprints, accelerating the shift from item-by-item audit to heuristic reliance (§7.8). The operating pressure of the platforms themselves—what cost constraints expanding submissions place on free operation—must be judged from each platform's cost and organizational material: one cannot infer from growing traffic that the platform must commercialize, nor take a single budget change as proof of the end of the academic commons; whether the pressure of the techno-capital clock forces preprint platforms toward productization is an empirical question requiring continuous tracking, not a structural conclusion that can be announced in advance.

§7.8 MCAF Accounting and Derived Indicators: The Cognitive Reynolds Number and the Narrative Leverage Ratio

The meta-formula of the Macro-Cognitive Accounting Framework (MCAF) is:

$$\Delta D_c = C \cdot \Lambda \cdot \phi - A \cdot E_c \cdot \|L\|.$$

The increment of cognitive debt (ΔD_c) is here an object of accounting, not a number obtained by directly subtracting several heterogeneous statistics. The products express the consequences of synergistic amplification and of the absence of key conditions; the minus sign requires both sides—expansion and verification—to be entered in the books. The six variables have the following observational counterparts in the academic scenario:

Variable	Observational counterpart in the academic scenario
Unlocked stock (C)	The existing literature, data, and methodological knowledge that get recalled and reorganized; its formation depends on prior long-term investment
Media conductivity coefficient (Λ)	How publication, retrieval, abstracting, citation, and evaluation systems deliver claims to users—the manner, speed, and selection rules
Narrative load (ϕ)	How journal prestige, indicator authority, and growth narratives obtain recognition for contributions still awaiting verification

Variable	Observational counterpart in the academic scenario
Meta-audit capacity (A)	Whether individuals, research communities, and institutions can trace grounds, examine methods, and question the tasks and boundaries of established evaluation
Human biological substrate (E_c)	The embodied constraints of reading, understanding, attention, and judgment; tools change work efficiency, but that does not mean humans can receive and verify without limit
Legitimacy base ($\|L\|$)	The institutional conditions supporting academic trust, the raising of objections, and the enforcement of responsibility—traceable reasons, procedural guarantees, and actual knowledge performance

This correspondence table does not permanently assign each activity to one party. Universities, journals, and researchers may all simultaneously amplify the reach of claims and organize their examination. Analysis must fix the concrete relations: who supplies what credentials to whom, who has the capacity to check, and who bears the consequences of not checking. Meta-audit also goes deeper than adding one more round of review: beyond checking whether a paper meets established requirements, it checks whether those requirements suffice to support the knowledge judgment being made.

After declaring the scenario boundaries, proxies, and aggregation rules, this chapter borrows two heuristic readings from Chapter 3 to examine information-arrival pressure and the support for narratives; they are not universal quantities derived algebraically from the meta-formula in the absence of scenario definitions (§0.5 has declared both to be domain instances of existing indicators of Chapter 3).

The cognitive Reynolds number ($Re_c \approx \Lambda/E_c$) can be used only for an individual scholar, a specified task, and a short slice of months to years: the unlocked stock, narrative load, meta-audit capacity, and legitimacy base may be held as background conditions and absorbed into the scenario scale; only then can it be written as $Re_c \approx \Lambda/E_c$ to compare information-arrival pressure per unit time with the capacity available for verification. Cross-period or macro-institutional analysis must return to the full relation between total projection load and total defensive capacity; global publication totals must not be treated as an individual scholar's information-arrival volume. Hanson et al.'s 47% serves only as evidence that the jointly indexed journal sample expanded; it neither estimates the cognitive Reynolds number nor measures arrival volume in an individual task. The reference magnitude for conscious information processing remains roughly 10 bits/s (Zheng & Meister, 2025; cited in Chapter 1 §1.4), but this book does not compute the denominator from that number. The ratio is used here only to ask an ordinal question within a specified task: is the volume of claims arriving growing faster than the verification effort actually invested?

The narrative leverage ratio ($\Omega = \phi/\|L\|$) serves here only as a qualitative question: can the narratives of journal prestige, indicator authority, and "quantifiable = scientific" exceed the verification supported by traceable reasons, procedural guarantees, and actual knowledge performance? Only after proxies, scales, and aggregation rules have been defined for both variables in a specific scenario can the analysis discuss relative

levels or critical intervals. Before that, "liquidation" must not be presented as a measured Ω crossing a bearing threshold. In §7.6, liquidation means the conversion of latent losses into explicit losses, not the completion of numerical calibration for the narrative leverage ratio.

§7.9 Counterfactual Examination: Is Pathological Recursion Structurally Inevitable?

The diagnosis, in the fiber framework, of the two counterfactual groups—historical and technological—of §3.7 in Part I is this: in the cases examined in this chapter, not a single case shows that pathological recursion has been reversed at the system level. The cases examined here show neither that the evaluation system has escaped pathological recursion nor that it is irreversible; the current state of the suppressing conditions must be traced case by case—"not reversed" does not entail "removed." Pathological recursion can appear as the distorted continuation of a functional circuit after its constraints change; this diagnosis does not presuppose a purely functional golden age. The task is to identify which conditions once kept the circuit answerable to epistemic scrutiny, when and how those conditions failed, and when the circuit's outputs began to replace independent verification.

Csiszar's diagnosis of the Cold War legacy (§3.1) receives a precise formulation in fiber language: if the "functional phase" was itself a construction under specific historical conditions, then diagnosing the current crisis as a "fall from function to pathology" presupposes a "golden age" that may never have existed. The correct diagnosis is: **pathological recursion is a structural tendency of indicator governance in the absence of constraints, not an accidental failure; it is suppressed under specific conditions and breaks out when those conditions fail—both suppression and outbreak are empirical questions.**

§8 How Publication Growth Became Proof of Institutional Success

§8.1 Decomposing Real Growth into Different Knowledge Processes

Imagine an evaluation system that has eliminated false authorship, fabricated data, fake reviews, and anomalous citations; the records in the database are accurate. This is a counterfactual set up for analysis. It removes the misconduct explanation, allowing the knowledge meaning of compliant growth to be examined separately.

Compare two paths that both increase publication volume. The first, through new observations, reliable re-examination, or the organization of material, excludes alternative explanations that previously remained viable, making clearer to those who follow when a conclusion can be used. The second divides existing work into more units, repeatedly extending expression across similar topics, adding publication records without excluding new explanations relevant to the original question. The difference here is not in the number of papers, nor in the use of new or old methods, but in what each production run leaves to the next round of research.

This is the positive criterion the fiber analysis yields: growth of knowledge can appear as the exclusion of important candidate explanations, the mapping of boundaries of applicability, or the acquisition of new evidence for questions that previously had no material to judge. Discovering that an old certainty lacks grounds, and reopening a question closed too early, are also knowledge progress—one cannot simply equate growth with the one-way shrinking of the candidate set. Correcting an error such that the number of papers supporting a theory decreases may equally add to knowledge. Eliminating fraud is therefore a real improvement, but only by continuing to discern these processes inside growth can we know which state of knowledge it improved.

The use of the impact factor also involves a whole-to-individual transfer. A journal's average citation performance, even if statistically accurate, is not sufficient to determine the actual influence of any one paper within it, still less its reliability (Kamat & Schatz, 2015). Repairing counting errors in journal indicators and examining individual papers are two adjacent but different tasks.

Three distinct diagnostic tasks follow: verifying research and records; limiting the range of interpretation of true readings; and tracking how evaluation changes the composition of output. The next step is not merely to admit that aggregate totals cannot answer all questions, but to explain why one kind of growth is more easily produced continuously within a given institution.

§8.2 How Compliant Production Organizes Itself around Evaluation

Researchers must often choose before results appear. A long-term question may require years of preserving materials, repeated failure, and the accumulation of methods; another topic can more easily yield a publishable result within the assessment deadline. If continued support depends mainly on recent publications, the two are not on equal footing in the competition for resources.

This mechanism presupposes neither that long-term questions are necessarily important nor that short-term research is necessarily shallow. What evaluation affects is which uncertain projects can continue to be examined. If there is no other way to recognize work that has not yet formed a paper, the system will tend to support paths that yield countable results earlier. The shortening of horizons and adjustment of publication strategies that Bornmann documents provide precisely the empirical entry into this selection process (Bornmann, 2011a).

The slicing of results requires the same discrimination. Complex collaboration, different questions, or improved methods can all constitute legitimate reasons for separate publication. The pathology is not that one piece of work yields several papers, but whether the slicing mainly adds to the count while damaging the integrity of argument, increasing the burden of duplication, or depriving the necessary synthetic interpretation of reward. Collyer's discussion of salami publication and career incentives concerns exactly this tension (Collyer, 2019).

The expansion of compliant production may therefore contain an institutional choice: work that fits the current indicators is more easily retained; work that does not fit gradually diminishes. It differs from violation, and it differs from researchers sincerely believing in the indicators. Even if participants retain their own quality judgments, the resource feedback may still require them to act by another set of standards.

After British evaluation shifted from quantity requirements to high-impact journals, adaptation shifted accordingly—showing that evaluation reform can change the direction of production, but may merely change the target of adaptation (Bornmann, 2011a). The interdisciplinary barriers Collyer discusses push this mechanism to the boundary of knowledge: contributions recognized by other disciplines may not be exchangeable, within one's own discipline, for the conditions of continued work (Collyer, 2019). What is selected is not only authors but which questions, materials, and interpretations can enter the next round of research.

A dynamic corollary follows: the output facing the next round of evaluation is not a sample formed independently of the previous round's rules. If a kind of work wins more easily, its share among surviving projects may rise; when evaluation subsequently observes more of this work, the change in composition cannot all be taken as evidence of its naturally higher knowledge value. Fiber analysis discerns the states compatible with the same reading; the analysis of dynamic corrosion goes further, showing how institutions change the opportunities for these states to be produced.

§8.3 From Output Growth to Rule Self-Certification

The circuit of institutional self-certification can be written as the following relation:

> Evaluation rules allocate resources → research organizes itself around the rules → countable output increases → growth is interpreted as success in knowledge production → the existing rules gain further support.

Two points in this circuit must be examined separately. The first is the production end: how the rules change research, and whether the change adds substantive contribution. The second is the certification end: by what the institution judges the change effective, and whether it merely recounts the output it has already rewarded. If production genuinely adds new material and effective examination, and certification can discern these increments, the circuit can support the accumulation of knowledge; if production concentrates on forms that win rewards easily, and certification examines only whether these forms have grown, the circuit can keep expanding without completing the verification of knowledge.

Espeland and Sauder's research on ranking reactivity shows that public measures change the activity of the measured (Espeland & Sauder, 2007; see §7.4). This chapter adds a further diagnosis: after rankings help shape the activity, the next round's positions first of all record relative position under the current round's rules; whether research or teaching tasks were better achieved requires different evidence. The same institution rising in several evaluations at once also requires tracking whether those evaluations reuse the same publication and prestige records. Multi-party recognition is multi-party calibration only when it adds effective constraints.

The "binding of the three lines" (§1.4) explains why this self-certification can persist. Researchers obtain professional credentials; universities obtain performance displays for funders; publication channels obtain stable demand—each party's local return can be real. As long as the same publication record can simultaneously satisfy these needs, local successes will support one another. The examination of knowledge may be borne by other people, at another time. This separation of cost and return lets production and certification keep running while errors are not yet visible or contributions not yet determined.

The circuit can also close along the basis of knowledge itself. Imagine an insufficiently verified judgment relayed by several reviews; later authors take these reviews as independent support for one another and continue to use the original judgment on the grounds that "a broad consensus already exists." What grows is the network of endorsements, not the underlying evidence; if the new adoptions are then used to prove the original judgment reliable, a pseudo-calibration loop has formed that runs without passing through resource allocation at all. This is a mechanism demonstration of input relations, not an empirical verdict on all reviews or citations.

A deeper feedback occurs in the conditions of calibration. When researchers re-check less because certification already exists, when less re-checking makes gaps in the certification less likely to surface, and when the absence of surfacing is read as the certification's reliability, the motivation to calibrate is weakened by the feedback itself; if evaluation simultaneously crowds out the work of forming new material and testing old conclusions, the independent evidence available to subsequent research also diminishes. Institutions do not merely block an already completed correction—they can weaken the conditions under which corrections arise. The preceding analysis of research selection has already touched this mechanism; §9 continues to examine how the speed gap between production and verification affects these conditions.

Institutional fulfillment and epistemic fulfillment can therefore come apart. A paper exchanged for a position, a

project passing its final review, a publishing service earning revenue—each has its own conditions of realization; whether a research claim can withstand examination, and whether existing results still permit correction, have another set of conditions. The framework's increment is to track, within the same historical chain, the separation of these conditions and its reproduction—not simply to declare that institutions pursued the wrong goals.

§8.4 What Citation Models Can Clarify: The Impact Parameter and Hot Streaks

Changing indicators is not meaningless work. A new indicator can repair specific defects of an old one and help make more appropriate comparisons. The problem is that the scope of the repair must remain clear: one cannot conclude, from a more complex mathematical form, that an independent judgment of knowledge has been obtained.

Sinatra et al. study the evolution of individual scientific impact, distinguishing in their model a relatively stable individual factor—the impact parameter (Q)—from paper-level randomness, and discuss the position of high-impact works within the publication sequence (Sinatra et al., 2016). This analysis helps separate cumulative citations, career stage, and the individual differences the model describes. It does not directly measure a scientist's judgment on concrete questions, nor can the model's parameters be read as an innate capacity unaffected by other conditions.

Liu et al.'s research on hot streaks in creative and scientific careers further distinguishes the concentrated emergence of high-impact work from contemporaneous changes in productivity (Liu et al., 2018). The position of a hot streak cannot simply be predicted from career age; its temporal clustering is not the same proposition. One cannot infer from the uncertainty of position that the works are fully independent of one another, nor infer from the absence of a productivity rise in some hot streaks that no effort to improve research conditions can work.

The significance of the two studies for evaluation is to make different temporal structures visible again, not to provide a final reading for judgment. Cumulative indicators easily conflate career duration with research contribution; impact models must themselves accept examination of their assumptions and data coverage. If they still process only citation records, they can discern new structure within the citation records, but cannot, on that basis alone, calibrate the true reliability of the cited work.

The critique of "using a clock to calibrate a clock" therefore has a clear boundary: it does not forbid developing new methods from old data, but refuses to take such a method's explanatory power over the old data directly as knowledge that new evidence would support. A better clock must still say what it has been aimed at.

§8.5 From Structural Early Warning to the Judgment of Concrete Loops

Chapter 4 of this book retains two diagnostic paths: retrospective tracking after failure has occurred, and structural early warning when failure has not yet surfaced. Academic evaluation likewise need not wait for fraud to be exposed or corrective findings to be institutionally disregarded before beginning to examine risk. When rewards concentrate on visible signals, when the bases of judgment largely overlap, and when independent verification lacks time and institutional support, there is already reason to protect the channels of examination. This is a judgment about conditions of risk, not a declaration that every result is defective.

The available materials support different links separately. Bornmann records publication adaptation after rule changes, showing that evaluation has entered production choices; Baccini et al.'s national comparisons make the institutional origin of citation records an object of explanation; Biagioli and Lippman show that impact signals

can be managed outside of content, showing that the certification infrastructure itself requires auditing; Collyer shows how publishability and disciplinary boundaries constrain research practice, showing that what is affected is not only the quality of finished products but the conditions under which knowledge is produced (Baccini et al., 2019; Biagioli & Lippman, 2020; Bornmann, 2011a; Collyer, 2019). These findings support the concrete channels of pathologization, not a single global proportion estimate.

To determine that some particular loop has entered pathological operation, the channels must be connected. For citation networks, track how managed signals return as grounds of authority; for research evaluation, track how output adapted to the rules obtains recognition again; for the use of knowledge, track whether repeated transmission from the same evidential source is taken as independent evidence, and whether objection can change the subsequent adoption of a claim. Whether resources are adjusted is one visible consequence; whether claims are revised, whether material is supplemented, whether calibration is restarted, are equally evidence of effective feedback.

Local correction and pathological feedback can coexist. A paper's correction shows that content-level correction worked at that point; if reviews, evaluations, or follow-up studies relying on the old version still relay the original conclusion as usual, the retraction of the error has not yet propagated along the relations of use. Conversely, even without a change in funding, if the correction is absorbed and research turns to more reliable grounds, repair at the level of knowledge has already occurred. Only so can repair be acknowledged and residue identified at once, rather than adjudicating the whole chain by a single resource outcome.

Retraction records should also be read this way. Proactively discovering problems can increase visible retractions while decreasing unaddressed risk; if a low retraction rate is directly rewarded as high quality, the exposure of problems may itself be punished. What the Macro-Cognitive Accounting Framework (MCAF) requires to be tracked is therefore not a prettier error statistic but the actual relations of discovery, handling, and cost-bearing.

The chapter accordingly makes a bounded judgment: in the commensurated evaluation discussed here, production adaptation and the management of certification signals have already brought about a discernible separation between institutional success and knowledge contribution; when these signals flow back as the basis of the next round of judgment without corresponding examination, pathological recursion has a path along which to run. The available materials suffice to identify and explain these paths, and to support a requirement for independent verification arranged before problems surface; they do not yet suffice to prove that all academic activity has been dominated by the same closed loop.

§9 Production, Dissemination, and Verification under AI Acceleration

§9.1 New Productive Capacity Enters Existing Incentives

Since 2022, generative applications of artificial intelligence (AI) have kept spreading, making text drafting, material organization, and formal processing easier to incorporate into automated workflows. But the expansion of academic production did not begin at this point. The 2016–2022 growth discussed by Hanson et al. already shows that the new tools entered a continuously expanding publication system, not an empty field where incentives had yet to form (Hanson et al., 2024).

This section does not estimate new knowledge from file volumes, model output, or cumulative platform submissions. Files can grow through images, attachments, and versions; a preprint and its journal version cannot

be counted indiscriminately as two independent discoveries. What is new must first be established before its relation to knowledge contribution can be discussed.

Generative tools can help express a real piece of research, and can also produce formally complete texts where no corresponding research exists. Using the same tool does not distinguish the two kinds of work. The former use may save labor and improve expression; the latter may lower the cost of manufacturing a credible appearance. Judging AI participation directly as low quality would bypass the needed judgment, just as judging a journal brand directly as high quality does.

What this section therefore offers is conditional mechanism analysis: after existing evaluation relations meet new productive capacity, which processes are more easily accelerated, and which verifications cannot be completed by the mere formation of text. On this basis it does not estimate the proportion of auto-generated papers in total publication, nor declare generative tools the sole cause of recent growth.

§9.2 Completed Text Is Not Completed Research

A well-structured paper can be formed quickly, but different research tasks have different conditions of completion. Empirical research must handle the relation between observations and experiments and claims; formal research must check derivations; historical and social research must account for materials, contexts, and competing interpretations. Writing these processes up as a paper is one link of the research work, not the sum of all links.

Bergstrom and West discuss the asymmetry between the low cost of manufacturing plausible content and the high cost of detecting its flaws (Bergstrom & West, 2020). Generative tools can amplify this asymmetry, but they can also help find inconsistencies, locate materials, or execute checks. One cannot first fix which side's capacity the tools raise, then write the conclusion up as a technological inevitability.

Three kinds of acceleration must be distinguished. Production acceleration reduces the labor needed to form a manuscript; dissemination acceleration makes manuscripts visible earlier and more widely; verification acceleration shortens, while preserving the force of examination, the process needed to exclude wrong explanations. An automatically generated checklist of methods belongs to the third kind only if the methods and materials have actually been checked; a form filled in faster shows only that the document was formed faster.

If assessment recognizes chiefly submittable output, the saved labor may be reinvested in more submissions; if evaluation recognizes the organization of primary material, independent replication, and synthetic interpretation, the savings may go to those activities instead. The same technical improvement can thus enter different feedback loops. What decides the outcome is not only what the tools can do, but what the institution continues to reward.

A producer–receiver asymmetry follows: one side saves writing time while the other side's load of claims to process may grow. As long as each individual verification still involves indispensable work, more submissions demand more verification effort—or push receivers toward greater reliance on filtering agents. The Macro-Cognitive Accounting Framework (MCAF) requires entering in the books at once the cost saved at the sending end and the burden added at the receiving end; taking generation efficiency as the efficiency of the whole knowledge system leaves out the second half of the ledger.

§9.3 How Production, Review, and Summarization Form Pseudo-Calibration

Imagine a workflow: a generative tool drafts a paper from a body of material; another system reviews the text; a third system summarizes the review and the paper; the summary is then used in the next round of production. The links have multiplied and the roles have grown, but this still does not determine how much independent verification has increased.

If the links mainly repeat the same body of erroneous material, and take linguistic completeness, formal regularity, and mutual consistency as grounds of endorsement, the new output can be ever smoother without excluding the shared false premises. The same error, returning through multiple links, easily appears with the look of multi-party endorsement. The problem here is the overlap of evidential sources, not a shortage of tools.

On the other hand, if an automated procedure actually checks verifiable derivations, or assists in executing experiments that can discriminate between alternative explanations, it cannot be excluded from calibration merely for using machines. A formal derivation passing inspection proves the conclusion under the corresponding formal conditions; it still does not automatically guarantee that the formalization accurately expresses the original question. Empirical data passing a format check is not the same as having verified the events the data record.

These two paths have different consequences in the next round. If review only confirms completeness of expression, subsequent production more easily takes that completeness as verified knowledge; if review finds an error in a derivation or counter-evidence in the raw material, the next round can revise the starting point. The same automated feedback can therefore either reinforce the appearance of certification or add effective constraints within the loop. The visible/hidden distinction thus not only explains what the tools miss; it also helps judge where to connect a tool so as to add knowledge.

The question of the human position arises here as well. If the author still chooses the question, traces the evidence, and handles counterexamples, the tools can share the labor; if the author hands over to the same automated certification chain even the judgment whether verification is needed, human presence does not equal the presence of calibration. Rather than drawing the line between human and machine, examine who retains contact with the original question, and who holds the right to stop the process and re-verify.

§9.4 The Opportunity of Open Dissemination and the Cost of Verification

Preprints allow results not yet certified by journals to enter discussion early. They increase the times and places where objection can arise, and let more researchers reach the material; this openness has real value (Harnad, 1990, 2008; Sever et al., 2019). But accessibility does not substitute for the reading, examination, and feedback that actually occur.

The other dimension of openness is the making itself. The readers of journal papers normally never see the referee reports; this arrangement was once defended by a proverbial saying whose earliest traceable form comes from the lawyer-poet John Godfrey Saxe's 1869 lecture (as transcribed in a newspaper report of the time):

> Laws, like sausages, cease to inspire respect in proportion as we know how they are made. (Saxe, 1869; as quoted in Shapiro, 2006)

John Baez (then professor of mathematics at the University of California, Riverside, and co-blogger at the n-Category Café, a group blog on math, physics, and philosophy), discussing the academic publishing crisis in 2010, noted that refereeing works exactly this way:

> the business of refereeing is kept invisible from the people who finally read the paper. (Baez, 2010)

In the same discussion, a participant described the experiment of the journal Atmospheric Chemistry and Physics: open access, refereeing that is by the referee's choice non-anonymous, supplemented by general public online discussion; non-anonymous reports were said to be of very high quality, and follow-up research even grew out of these and their public discussion. Baez then turned the proverb upside down:

> perhaps sausages will be better if the public is allowed to watch them being made. (Baez, 2010)

What comes after the inversion is the real question of this section: being able to watch the making is not the same as checking going on. Open referee reports still have to be read and tested; what is said below about filtering and brands applies equally to openly available refereeing material.

When candidate material multiplies, researchers must decide within limited time what to read and what to re-examine. Search, filtering, and summarization tools can help make this decision, and will also affect which material is more easily noticed. If the main basis of filtering remains brand, citations, and familiar forms of expression, open dissemination does not necessarily give previously neglected contributions their chance at evaluation.

Rosa's distinctions within social acceleration help separate technical speed from the organization of human work: faster transmission does not automatically accelerate at the same rate the activities that require digestion, discussion, and long-term observation (Rosa, 2013). If claims to be processed grow faster than actual verification capacity, maintaining the same verification requirements demands stricter filtering or longer waiting; if verification capacity is not increased while adoption expands with output, the system more easily expands its use of insufficiently examined material. The rise of media conductivity and the insufficiency of meta-audit capacity can then occur together: the same open system improves conditions in the dimension of access while still requiring new investment in the dimension of verification.

This is not an inference from one person's processing ceiling to the inevitable failure of the whole institution. Institutions can divide the labor of checking, share materials, and preserve version and correction relations; they can also change arrangements that reward only new claims. What must be compared is the arrival of claims in a specific task, the actual verification effort, and the results—not the global paper count divided by some individual's cognitive rate.

Preprint platforms themselves also require continuous operation, maintenance, and moderation; their costs and organizational conditions should be assessed separately within the evidential limits set out in §7.7.

§9.5 Filtering Cannot Bring Back Research That Was Never Done

What changes with production incentives is not only the quality of already formed papers, but whether material has the chance to form at all. Long-term investigations, the preservation of failed experiments, the organization of primary materials, and cross-domain synthesis often do not correspond one-to-one with short-cycle publication. If these activities diminish for lack of continued support, later cleanups of unreliable texts cannot bring them back.

This point goes deeper than identifying generated text. Even if every manuscript in the future could accurately declare its mode of production, the label still would not say which important observations were never made, and which questions were abandoned before they could form publishable results. Judgments about missing material

of course also require evidence—one cannot imagine every unperformed study as a lost major contribution; project records, funding arrangements, and comparisons of research processes are what may help discern them.

The risks under AI acceleration therefore have two levels: whether existing material can be reliably verified, and whether the institution continues to support the formation of new evidence. The former can begin with checking existing papers; the latter requires tracking the conditions of production. Filtering can reduce the entry of errors into subsequent use; it cannot substitute for the work of maintaining independent observation and research.

This distinction also keeps a question open for the study of platforms later in the book: what information systems inherit is not only text, but which records once had the conditions to be produced. What academia offers is a trackable institutional case, not a pre-issued verdict that all training material has already gone bad.

§10 Disciplines, Institutions, and the Differences of Reform

§10.1 Disciplines Are Not Homogeneous Evaluation Environments

Different disciplines relate differently to "publish or perish." The single monograph of the humanities and social sciences, the series of papers in the natural sciences and engineering, the multicenter trial report of clinical medicine—each has formed its own publication rhythm, authorship norms, and citation culture. The same indicators (impact factor, h-index, citation counts) carry vastly different meanings across disciplines—§3.5 has described the Web of Science's coverage bias and low counts for the humanities and social sciences (Almeida et al., 2018): to evaluate across disciplines with a single ruler is in effect to take one discipline's publication culture as the standard of all.

§10.2 Institutional Environments Shape How Indicators Operate

The same indicator operates differently in different institutional environments. Britain's RAE/REF is a nationally organized periodic review; evaluation in the United States is dispersed across universities and funding agencies; in Germany's traditional chair system, full professorships are few and highly autonomous, with large numbers of researchers working under chairs on fixed-term contracts, the professorship itself a scarce signal (Schwabe et al., 2024); China's excellence initiatives wire international rankings directly into resource allocation (detailed in §4). How an indicator is wired into these institutions determines the concrete shape of its being gamed—Baccini et al.'s analysis of Italy's ANVUR reform is one instance (Baccini et al., 2019; see §2.3).

§10.3 "Unquantifiability" Is Not a Ground for Declining Explanation

The critique of commensuration does not directly yield "academic work is not to be accounted for." The humanities' and social sciences' resistance to quantitative evaluation has two different versions: one holds that academic judgment rests on substantive understanding that cannot be compressed into numbers—this is legitimate; the other takes "unquantifiability" as a ground for refusing to give an account of one's work to anyone—this confuses "cannot be precisely measured" with "need not account for itself." Zhao Dingxin's diagnosis of the ills of the interpretive tradition applies here: researchers who "use rhetorical splendor and theoretical obscurity to conceal the poverty and emptiness of their content" (Zhao, 2004; author's translation)—the critique targets the substitution of an appearance of profundity for the substantive duty of explanation, and it applies equally to appeals to "essential complexity" as a way of avoiding explanation. Judgment cannot be replaced by indicators; but judgment must be able to give reasons.

§10.4 What Reforms Add in Discriminative Capacity

The judgment of reforms can proceed along the distinctions above, without first sorting every proposal into success and failure.

Reform or check	Link acted on directly	Basis for judging the actual increment	Conditions still needing maintenance
Cleaning duplicate records, false authorship, anomalous citations	Counts and impact signals	Erroneous records corrected; distorted impact credentials no longer carried forward	Signal repair and examination of research content proceed separately
Open access, preprints, self-archiving	Access conditions and the order of dissemination	More people can reach the material, raise objections, and track responses	Actual reading, verification, and platform maintenance have someone to bear them
Adding evaluative dimensions, improving bibliometric models	Information retention and modes of comparison	Key differences previously merged become discernible	The new indicator's task made explicit; readings from the same source not taken as independent support
Supporting replication, negative results, preservation of primary material	The basis of knowledge and calibration	Conclusions and boundaries of applicability corrected; new evidence enters subsequent use	Re-examiners can keep working; corrections propagate along dependency relations
Stable support for long-term research	The formation of new material and new questions	Necessary work that cannot yield papers within the assessment period continues and remains open to examination	Professional review tracks actual progress, not slowness substituted for quality
Adjusting indicator thresholds in hiring and funding	The institutional force of judgment	Traceable reasons beyond the indicator can change decisions	Conflicts of interest, grounds of discretion, and appeals remain subject to review

This table is a set of analytical criteria, not a statistical conclusion that the various reforms already work everywhere. Cleaning counts has its own value, and cannot be denied because it does not repair all of evaluation; open dissemination need not carry the whole burden of abolishing professional competition. But local functions cannot redeem the institution's knowledge promise as a whole.

These reforms also form paths that can reinforce one another. Open material makes re-examination more feasible; re-examination makes errors and boundaries of applicability visible; correction relations let subsequent use update itself; and evaluation rules then provide continuing conditions for this work. Each link in the chain has independent value; combined, they may improve both current judgment and the next round's capacity for calibration.

Still to be examined separately: whether problems are discovered, whether discovery can change decisions, and whether anyone continually bears the work of verification. Knowing that a study has a problem, yet lacking a procedure by which the finding changes decisions—the lack is in the institutional force of judgment; allowing objections, yet no one having time to obtain the material—the lack is in the conditions of work. Granting reviewers discretion without requiring checkable reasons may equally restore private bias. Preserving judgment therefore includes both letting judgment occur and keeping judgment itself subject to review.

§10.5 Long-Term Support, Slow Science, and the Conditions of Change

Whitley distinguishes the British organization of 1950–1980, which relied more on researchers' discretion, from the later arrangements allocating resources more through research evaluation (Whitley, 2010; cited in §3.7). The comparison makes continued support and professional discretion variable institutional conditions, rather than a space scholars can only fight for by personal will. The advance of research evaluation after 1986 shows in turn that these conditions shift with public governance; that a larger space of autonomy once existed proves neither that it was free of power relations nor that it was later necessarily replaced through internal failure.

What slow science emphasizes—thinking, digestion, long-term work—concerns precisely the investments that short-cycle assessment cannot recognize. Gingras's critique of evaluation demands discerning the relation between indicators and research tasks, not substituting slowness for speed and founding another standard of quality that needs no explanation (Gingras, 2016).

The value of long-term support lies in letting work that cannot yet prove itself through short-term publication continue to receive substantive examination. It still requires accounting for research progress, the state of the material, and how questions change. Automatically equating the absence of papers with profundity commits the same interpretive overreach as automatically equating many papers with excellence.

Whether such arrangements persist also depends on how funding, hiring, and external accountability fit together. An individual slowing down while still competing at the same publication threshold may merely concentrate the cost on the individual. Local reforms that take effect may also be limited by changes in external conditions. The latter limitation should enter the judgment of effect, without erasing the improvements that have locally occurred.

Hendricks's analysis of coordination failure shows the distance between recognizing a problem and collectively changing behavior; Beigel et al.'s discussion of the loss of money, time, trust, and control reminds us that the costs of reform are borne by different participants (Beigel et al., 2025; Hendricks, 2026; their "sucker/hustler" distinction was cited in §2.3). Institutional change must handle these costs, not explain the difficulty away as everyone having lost the will to judge.

This section and §0.4 above differ in the level of their conclusions: §0.4 concerns the difficulty of retreat as a synthesis of mechanisms; this section concerns an empirical judgment at the system level—that institutional mechanisms make reversal difficult does not mean it is empirically absolutely irreversible. That history cannot return unchanged, that institutions have path dependence, that restoring research conditions costs something, are all debatable judgments; together they do not automatically prove absolute irreversibility. As long as counter-evidence, correction, and different reasons of evaluation can still gain practical force, the loop has parts that can be changed and deserve careful discernment.

Part IV: Self-Reference and Conclusion

§11 Self-Reference: This Chapter as a Mapping Operation

This chapter is itself a mapping operation—it "projects" insights scattered across multiple disciplines into readings of the fiber framework. If, after finishing it, the reader feels he has "understood" the concept of the fiber, he should ask: does this understanding come from calibration (he has independently tested the argument against his own experience), or from another reading (he has been persuaded by the argument's narrative)? The latter would be one more instance of closed-loop pseudo-calibration.

The baroque self-check applies equally to this chapter. The chapter retains ten concept entries, five independently load-bearing (the self-referential audit at the opening of §5), the rest auxiliary levels. But is the deletion test itself also a mapping operation—projecting the question "does the concept bear load" onto a binary verdict, a verdict that may itself carry the author's subjective bias? The honest answer is: yes. The deletion test is a lossy mapping, but its alternative—not running the deletion test—would not remove the lossiness, only make it invisible. Marking the lossiness is safer than pretending to losslessness. Von Neumann's reminder that mathematics tends toward the baroque can serve here as self-check: growing refinement of form does not guarantee continued contact with the original question (von Neumann, 1947). The value of a concept is proven not by its number, name, or the appearance of its formula, but tested by which step of discrimination it actually accomplishes.

The chapter's several judgments therefore carry different evidential burdens. That the same reading cannot distinguish two constructed candidate states is a logical judgment under given conditions; that some evaluation causes shifts in topic selection requires empirical material; to judge pathological recursion, one must show how internal output flows back, what independent grounds it replaces, and how feedback amplifies bias or further erodes the conditions for calibration. Identifying the risk conditions can support early warning; asserting that some concrete loop already runs this way requires corresponding process evidence. These levels cannot be merged merely because they share the same vocabulary.

Gap one: the Goodhartization risk of "annotating the fiber" once it is itself metricized. The practical import of §12 is "annotating the fiber"—labeling, beside every reading, the space of possibilities behind it. But if "has annotated the fiber" becomes a new evaluation standard, will it not become a new pseudo-small fiber? "This study annotated the fiber" → "studies that annotate the fiber are high-quality studies" → "researchers have an incentive to annotate the fiber formally without substantively changing practice" → "annotation" itself is Goodharted. The honest answer is: the risk exists; annotating the fiber is not immune to dynamic corrosion—any indicator used for resource allocation can face gaming pressure; whether gaming occurs and what form it takes depend on actors' room for adjustment, the reward structure, and independent conditions of verification. The mitigation is not to abandon annotation (which would only make fibers invisible) but to embed in the annotation a second-order annotation that "this annotation itself may be gamed"—which leads to infinite regress. The framework's honest boundary here is: on the premise that the limitations are real, annotating the fiber is better than not annotating; but annotation is not an endpoint; it is the starting point of continuous audit. The practical meaning of annotation is to state which important states a reading is still compatible with, which interpretations remain unexcluded, and what verification is still needed—it is not the addition to the end of a paper of a universally applicable limitation statement, still less a license to credibility earned merely by the presence of uncertainty words. One should check whether the limitations actually change conclusions: does an

as-yet undiscerned difference require suspending the original ordering; can a clear piece of counter-evidence cause an existing endorsement to be reconsidered.

Gap two: the fiber framework's terminology itself as narrative load. §5.7 diagnoses "narrative load as the narrative packaging projected by the projecting party in pursuit of a local optimum"—is the fiber framework's own terminology not also narrative load? The reader persuaded by "fiber/pseudo-small fiber/dynamic corrosion" forms the perception of having "understood"—whether this perception is reliable cannot be settled by its persuasiveness: the precise mathematical appearance ($\pi^{-1}(r)$, Re_c, Ω) may make the account appear more rigorous than its definitions and evidence warrant (the lack of operational definitions for the derived indicators, the limits of the analogies, the absence of falsifiable predictions). The honest answer is: the fiber framework's terminology does carry narrative load—the mathematical lineage of the word "fiber" suggests a depth of mathematical rigor greater than what is actually used. The lineage statement at the opening of §6 is the response to this gap—acknowledging the gap between attribution of lineage and mathematical kernel is the first step in making the narrative load auditable. Likewise, preserving judgment does not mean letting the judgment of a few go unexamined: Weizenbaum's distinction between judgment and calculation reminds us that calculation cannot carry unexplained value choices, but it grants no natural correctness to any individual intuition (Weizenbaum, 1976). Judgment needs reasons, material, professional division of labor, and arrangements open to objection, just as indicators need to account for their computation and conditions of applicability.

Gap three: "ten items" as an instance of the impulse to count. §7.3 of this chapter criticizes the "salami slicing" strategy described by Bornmann—cutting research into countable thin slices to fit the indicator. Yet §5 gives the precisely countable list of "ten concept entries," in self-referential tension with the behavioral pattern the chapter criticizes. The self-referential audit at the opening of §5 has responded to this gap—acknowledging that "ten" bears no load, that the independently load-bearing core is five concepts, and that the ten-item presentation is kept for the trackability of the argument, not for conceptual independence. Is this response sufficient? Not entirely—acknowledging the impulse to count is not eliminating it. The more thorough treatment would reorganize §5 from "a parallel list of ten items" into a hierarchy of "five core concepts plus auxiliary concepts marked as mergeable or derivable." The chapter retains the current structure as a pragmatic trade-off, but the reader should know that this is an unfinished self-referential audit.

Counterevidence must also be genuinely effective. If a reform only cleans citations yet improves research reliability by releasing verification resources, one cannot insist that it repairs only counting; if several sets of tools do add independent examination, one cannot convict them of calibrating clocks with clocks merely because they all use numbers. New evidence should change the judgment of concrete processes, not always be absorbed by the framework as one more illustration.

§12 Conclusion: Annotating the Fiber

The core proposition of the fiber framework can be put in one sentence: **every observation has an inverse image under a given mapping; when that inverse image contains multiple states that are not equivalent for the question at issue, the observation leaves uncertainty.** For a non-singleton fiber, the same reading and its post-processing cannot eliminate the indistinguishability among those states; when the mapping is used for decisions, the empirical relation between readings and the evaluative target may be rewritten by use (dynamic corrosion); when optimization is driven only by preference signals and lacks independent calibration, the system

may manufacture pseudo-small fibers—making large fibers look small.

The practical import of the fiber framework is not to "eliminate every fiber" or to "turn every reading into a singleton," but to **annotate the fiber**: beside every reading, honestly labeling "how large is the space of possibilities behind this reading," "what factors this mapping has ignored," "whether the system's tone of certainty matches its real uncertainty." Annotating the fiber does not eliminate uncertainty; it makes uncertainty visible. Manski reporting identification regions when the evidence permits only partial identification (Manski, 2003), Vovk et al. generating prediction sets under stated coverage conditions (Vovk et al., 2005)—these are engineering references for the "annotating the fiber" path in their respective fields. The practical effect of annotation can occur at different positions: letting readers go back to the primary material, letting researchers revise claims, letting reviewers suspend a comparison whose basis is insufficient, or letting institutions support new verification. It need not first change funding to be meaningful; but neither can readers having been reminded be taken as the duty having been discharged.

For the "publish or perish" system, annotating the fiber means: beside every impact factor, labeling "how large is the fiber of the mapping between this number and academic quality"; beside every ranking, labeling "which unquantifiable dimensions this sequence conceals"; beside every h-index, labeling "how many-to-many is the relation between citation counts and academic contribution." Forgetting that the fiber exists is more dangerous than the fiber itself.

A thousand clocks sharing the same bias and bringing no new constraints cannot answer "what time is it" by agreeing with one another. A thousand papers cannot answer "what is true." A thousand indicators cannot answer "how good." Not because there are not enough of them—because such multitude is a multitude in the space of mappings, not in the space of truth; number adds to the grounds of judgment only when it brings new effective constraints.

One historical achievement of the academic publication system is that claims can be preserved, disseminated, and subjected to the examination of those who come later beyond private exchanges. Professional support lets research continue; commercial operation and digital infrastructure have enlarged the organizational capacity of dissemination; at the same time, the publication record has been converted into the credential of positions, institutional performance, and operating income. That the same record serves different uses is the organizational condition of the accumulation of knowledge, and also the seam where the knowledge function may be squeezed by other goals.

The chapter's first corollary is that the knowledge meaning of publication growth must be discerned from the research processes within it. New material, effective counter-evidence, the clarification of boundaries of applicability, and new questions can all add to knowledge; a true count can only certify the counted activity, not testify for these processes. The fiber analysis accordingly separates misconduct, interpretive overreach, and changes in the composition of compliant production, so that evaluation need not wait for the discovery of fraud before judgment restarts.

The second corollary is that evaluation not only selects research but participates in forming the material by which it is later evaluated. Research adapts to the rules; the adapted output then supports the rules—constituting the institutional path of growth's self-certification; existing claims gain more recognition through repeated transmission from the same evidential source, and the recognition is then used to support continued adoption—constituting the feedback inside the basis of knowledge. The pathology lies not in the existence of loops but

in the loops' replacing the needed independent constraints with internal readings, and continuously amplifying bias or eroding the conditions of calibration. Resource feedback is one key channel among them; it does not exhaust the process.

The third corollary is that the capacity of calibration is itself a condition that knowledge production must reproduce. Only by making material accessible, keeping proofs and explanations checkable, letting counter-evidence change subsequent use, and preserving time and institutional force for this work can one round of results assist the next round of research. If only output is rewarded while these conditions are continuously consumed, the institution can grow in publications and revenue while accumulating cognitive debt. Generative tools push this separation to new conditions of cost and speed: the gains of production efficiency must be settled jointly with the burden of verification and the conditions for forming new evidence.

These corollaries do not require a return to some golden age. The public record, preprints, independent re-examination, and professional discretion all provide conditions that can be maintained and recombined, each of whose effects must be discerned item by item. The knowledge function of the academic community is not to keep the clocks consistent with one another, but to keep material, questions, and counter-evidence binding on judgment, and to keep cultivating people able to use these constraints. An evaluation system cannot prove itself effective merely by the growth of the output it rewards; growth acquires its knowledge meaning only by continuing to support this work.

§12.1 The Framework's Calibration Case and Unfolding Cases

The diagnostic scope of the fiber framework is not limited to academic publishing. This chapter chose the academic publishing system as its calibration case on a methodological judgment: academic publishing is a richly documented analytical environment—institutional boundaries clear, history traceable over three and a half centuries (from the founding of the journals in 1665 to the present), rich traces connecting publication records and evaluation rules. The analytical tools explicated and examined through process evidence in this setting (fiber, dynamic corrosion, and cognitive debt), together with the three clocks, the cognitive Reynolds number, and the narrative leverage ratio retained as heuristic mappings, may be carried into structurally more complex environments only after their observational objects, proxies, and conditions of application have been redefined and retested.

The digital platform ecosystem—the "Discrete Leviathan" series of later chapters of this book—is the framework's unfolding case. The two systems share the same deep mechanism: a continuous, hard-to-observe quantity is compressed into countable discrete signals; the signals are used for institutional decisions; when actors can adjust around those signals and the reward structure supplies corresponding incentives, the signals can face gaming pressure, their empirical relation to the original quantity can change, and the consequences can be delayed. In academic publishing this "discretization → dynamic corrosion → delayed manifestation" chain appears as: academic quality → impact factor/citation counts → used in hiring/funding decisions → conditions for dynamic corrosion established → reproducibility problems surfacing after extended delays in some research and use relations. In digital platforms the same chain appears as: human behavior/attention → clicks/dwell/completion rate → used in ad targeting/content distribution decisions → engagement indicators maximized → expression compressed into algorithmically recognizable formats → the slow effects of attention fragmentation and cognitive degradation.

The revenue chain of academic publishing analyzed in this chapter runs mainly through subscription fees, APCs, and institutional purchasing—this does not mean academic journals carry no advertising; the dominance of targeted advertising auctions in the revenue structure is the key condition distinguishing the unfolding case from the calibration case. The monetization logic of digital platforms is attention → behavioral signals → advertising's targeted auctions—advertising is the engine. The two systems differ in their principal revenue mechanisms, but both can discretize inputs, processes, and outputs—academic metrics assess results and also reshape research choices (§8.2); platform metrics record user behavior and influence content production and distribution. The comparison concerns how particular readings acquire commercial and decision-making authority.

A gray zone must be marked. The impact factor itself performs an advertising-like function of packaging: packaging "possibly good" as "certainly good" ("advertising" here is an analogy, not a reference to the sale of ad space or targeted auctioning), building confidence not grounded in substance before information-disadvantaged buyers (libraries, hiring committees, funding agencies). Garfield and Irving H. Sher created the impact factor in the early 1960s for the journal selection of the Science Citation Index (Garfield, 2005), and the 1972 paper carried it into journal evaluation—the impact factor was from birth a marriage of evaluation service and commercial product. The more recent trend is altmetrics—reading the traces of dissemination across news, policy documents, and social media as a paper's "impact"—and the conflation of attention with value occurs only when evaluation systems treat these attention readings as proof of quality.

The practical import of this bridge is: when readers carry the fiber framework used in this chapter for case-based explication and process examination into the Discrete Leviathan series, they should attend to where the advertising variable enters, how it changes the site where discretization acts, and where the two systems' mechanisms of dynamic corrosion and delayed manifestation are isomorphic and where they diverge. The framework's tools are general; but every case requires recalibrating the concrete structure of the mapping function.

References

Agarwal, A., & Ioannidis, J. P. A. (2019). PREDIMED Trial of Mediterranean Diet: Retracted, Republished, Still Trusted? *BMJ*, 364, l341. https://doi.org/10.1136/bmj.l341

Agrawal, A. (2025). Country-Wise Retraction Analysis from 2022–2024: Increased Publishing Leading to Higher Retraction Rates. Zenodo. https://doi.org/10.5281/zenodo.14634373 [Preprint, not yet peer-reviewed]

Almeida, I. C. de S., Almeida, R. G. de, & Carvalho, L. R. de. (2018). Academic Rankings and Pluralism: The Case of Brazil and the New Version of Qualis. *EconomiA*, 19(3). https://doi.org/10.1016/j.econ.2018.03.003

Baccini, A., De Nicolao, G., & Petrovich, E. (2019). Citation Gaming Induced by Bibliometric Evaluation: A Country-Level Comparative Analysis. *PLOS ONE*, 14(9), e0221212. https://doi.org/10.1371/journal.pone.0221212

Baez, J. (2010). Boycotting the Nature Publishing Group. *The n-Category Café*, June 9. https://golem.ph.utexas.edu/category/2010/06/boycotting_the_nature_publishi.html

Baldwin, M. (2015). Credibility, Peer Review, and Nature, 1945–1990. *Notes and Records of the Royal Society of London*, 69(3). https://doi.org/10.1098/rsnr.2015.0029

Begley, C. G., & Ioannidis, J. P. A. (2015). Reproducibility in Science: Improving the Standard for Basic and

Preclinical Research. *Circulation Research*, 116. https://doi.org/10.1161/CIRCRESAHA.114.303819

Beigel, F., et al. (2025). The Drain of Scientific Publishing. arXiv:2511.04820. https://arxiv.org/abs/2511.04820 [Preprint; a public review record is also available]

Bergstrom, C. T., & West, J. D. (2020). *Calling Bullshit: The Art of Skepticism in a Data-Driven World*. New York: Random House. Chinese edition: Hu Xiaorui, trans., *Calling Bullshit*. Beijing: CITIC Press Group, 2022. (In Chinese.)

Berners-Lee, T., et al. (2006). A Framework for Web Science. *Foundations and Trends in Web Science*, 1(1). https://doi.org/10.1561/1800000001

Biagioli, M., & Lippman, A. (Eds.). (2020). *Gaming the Metrics: Misconduct and Manipulation in Academic Research*. Cambridge, MA: MIT Press.

Bornmann, L. (2011a). Mimicry in Science? *Scientometrics*, 86. https://doi.org/10.1007/s11192-010-0222-8

Bornmann, L. (2011b). Scientific Peer Review. *Annual Review of Information Science and Technology*, 45. https://doi.org/10.1002/aris.2011.1440450112

Bornmann, L., & Mutz, R. (2015). Growth Rates of Modern Science: A Bibliometric Analysis Based on the Number of Publications and Cited References. *Journal of the Association for Information Science and Technology*, 66(11). https://doi.org/10.1002/asi.23329

Burawoy, M. (2005). For Public Sociology. *American Sociological Review*, 70(1). https://www.jstor.org/stable/4145348

Burke, P. (2000). *A Social History of Knowledge: From Gutenberg to Diderot*. Polity Press.

Burke, P. (2012). *A Social History of Knowledge II: From the Encyclopédie to Wikipedia*. Polity Press.

Campbell, D. T. (1979). Assessing the Impact of Planned Social Change. *Evaluation and Program Planning*, 2(1). https://doi.org/10.1016/0149-7189(79)90048-X

Catanzaro, M. (2024). Citation Cartels Help Some Mathematicians—and Their Universities—Climb the Rankings. *Science*, 383(6682). https://doi.org/10.1126/science.zcl2s6d

Collingridge, D. (1980). *The Social Control of Technology*. London: Frances Pinter.

Collyer, T. A. (2019). 'Salami Slicing' Helps Careers but Harms Science. *Nature Human Behaviour*, 3(10). https://doi.org/10.1038/s41562-019-0687-2

Csiszar, A. (2018). *The Scientific Journal: Authorship and the Politics of Knowledge in the Nineteenth Century*. Chicago: University of Chicago Press.

Ding, Y. (2001). In China, Publish or Perish Is Becoming the New Reality. *Science*, 291(5508). https://doi.org/10.1126/science.291.5508.1477

Erren, T. C., Shaw, D. M., & Morfeld, P. (2015). Analyzing the Publish-or-Perish Paradigm with Game Theory: The Prisoner's Dilemma and a Possible Escape. *Science and Engineering Ethics*, 22(5). https://doi.org/10.1007/s11948-015-9701-x

Espeland, W. N., & Sauder, M. (2007). Rankings and Reactivity: How Public Measures Recreate Social Worlds. *American Journal of Sociology*, 113(1). https://doi.org/10.1086/517897

EUA (2018). *The Lack of Transparency and Competition in the Academic Publishing Market in Europe and Beyond*. https://www.eua.eu/images/publications/The_lack_of_transparency_and_competition_in_the_academic_publishing_market_in_Europe_and_beyond.pdf

Fyfe, A., et al. (2017). *Untangling Academic Publishing: A History of the Relationship Between Commercial Interests, Academic Prestige and the Circulation of Research*. Zenodo. https://doi.org/10.5281/zenodo.546100

Fyfe, A., et al. (2022). *A History of Scientific Journals: Publishing at the Royal Society, 1665–2015*. UCL Press. https://doi.org/10.14324/111.9781800082328

Gagnon, S. (2017). Journal Publishers' Big Deals: Are They Worth It? *Against the Grain*, 29(2), Article 48. https://doi.org/10.7771/2380-176X.7747

Garfield, E. (2005). The Agony and the Ecstasy—The History and Meaning of the Journal Impact Factor. Paper presented at the International Congress on Peer Review and Biomedical Publication, Chicago, IL, September 16. https://garfield.library.upenn.edu/papers/jifchicago2005.pdf

Gingras, Y. (2016). *Bibliometrics and Research Evaluation: Uses and Abuses*. Cambridge, MA: MIT Press.

Gingras, Y. (2020). The Transformation of the Scientific Paper: From Knowledge to Accounting Unit. In M. Biagioli & A. Lippman (Eds.), *Gaming the Metrics: Misconduct and Manipulation in Academic Research*. MIT Press.

Goodhart, C. A. E. (1975/1984). Problems of Monetary Management: The U.K. Experience. In *Papers in Monetary Economics*, Vol. I, Reserve Bank of Australia (1975); reprinted in *Monetary Theory and Practice* (1984).

Gordon, G. (2017). Indicators, Rankings and the Political Economy of Academic Production in International Law. *Leiden Journal of International Law*, 30(2). https://doi.org/10.1017/S0922156517000188

Grudniewicz, A., et al. (2019). Predatory Journals: No Definition, No Defence. *Nature*, 576(7786). https://doi.org/10.1038/d41586-019-03759-y

Hanson, M. A., et al. (2024). The Strain on Scientific Publishing. *Quantitative Science Studies*, 5(4). https://doi.org/10.1162/qss_a_00327

Harnad, S. (1990). Scholarly Skywriting and the Prepublication Continuum of Scientific Inquiry. *Psychological Science*, 1(6). https://www.jstor.org/stable/40062820

Harnad, S. (2006). Publish or Perish—Self-Archive to Flourish: The Green Route to Open Access. *ERCIM News*, 64. https://ercim.eu/publication/Ercim_News/enw64/harnad.html

Harnad, S. (2008). Waking OA's 'Slumbering Giant': The University's Mandate to Mandate Open Access. *New Review of Information Networking*, 14. https://doi.org/10.1080/13614570903001322

Hendricks, V. F. (2026). *Sucker Nation: The Philosophy of Trading Wits for Vanity*. Springer. https://doi.org/10.1007/978-3-032-22788-1

Hirsch, J. E. (2005). An Index to Quantify an Individual's Scientific Research Output. *PNAS*, 102(46). https://doi.org/10.1073/pnas.0507655102

Holst, V., et al. (2026). Dataset Artefacts Can Partially Drive the Measured Decline in Disruption. *Nature*, 656(8127). https://doi.org/10.1038/s41586-026-10787-y

Ioannidis, J. P. A., Boyack, K. W., & Klavans, R. (2014). Estimates of the Continuously Publishing Core in the Scientific Workforce. *PLOS ONE*, 9(7), e101698. https://doi.org/10.1371/journal.pone.0101698

Jefferson, T., et al. (2007). Editorial Peer Review for Improving the Quality of Reports of Biomedical Studies. *Cochrane Database of Systematic Reviews*, 2007(2), MR000016. https://doi.org/10.1002/14651858.MR000016.pub3

Kamat, P. V., & Schatz, G. C. (2015). Journal Impact Factor and the Real Impact of Your Paper. *The Journal of Physical Chemistry Letters*, 6(15). https://doi.org/10.1021/acs.jpclett.5b01527

Knoche, M. (2025). *Critique of the Political Economy of the Media: Foundations and Applications*. University of Westminster Press. https://doi.org/10.16997/book79

Krantz, D. H., et al. (1971). *Foundations of Measurement, Vol. I: Additive and Polynomial Representations*. New York: Academic Press.

Kronick, D. A. (1976). *A History of Scientific and Technical Periodicals: The Origins and Development of the Scientific and Technical Press, 1665–1790* (2nd ed.). The Scarecrow Press.

Larivière, V., et al. (2014). arXiv E-prints and the Journal of Record: An Analysis of Roles and Relationships. *Journal of the Association for Information Science and Technology*, 65(6). https://doi.org/10.1002/asi.23044

Larivière, V., Haustein, S., & Mongeon, P. (2015). The Oligopoly of Academic Publishers in the Digital Era. *PLOS ONE*, 10(6), e0127502. https://doi.org/10.1371/journal.pone.0127502

Liu, L., et al. (2018). Hot Streaks in Artistic, Cultural, and Scientific Careers. *Nature*, 559. https://doi.org/10.1038/s41586-018-0315-8

Macdonald, S. (2025). The Scandal of Academic Publishing. *Publishing Research Quarterly*, 41. https://doi.org/10.1007/s12109-025-10042-8

Mandelbrot, B. B. (2012). *The Fractalist: Memoir of a Scientific Maverick*. New York: Pantheon.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. New York: Springer. Chinese edition: Wang Zhongyu, trans., *Partial Identification of Probability Distributions*. Harbin: Harbin Institute of Technology Press, 2018. (In Chinese.)

Martin, B., & Whitley, R. (2010). The UK Research Assessment Exercise: A Case of Regulatory Capture? In R. Whitley, J. Gläser, & L. Engwall (Eds.), *Reconfiguring Knowledge Production: Changing Authority Relationships in the Sciences and Their Consequences for Intellectual Innovation*. Oxford: Oxford University Press.

McGilchrist, I. (2021). *The Matter with Things: Our Brains, Our Delusions and the Unmaking of the World*. London: Perspectiva Press.

Merton, R. K. (1973). *The Sociology of Science: Theoretical and Empirical Investigations* (N. W. Storer, Ed.). University of Chicago Press.

Ministry of Education, & Ministry of Science and Technology. (2020). Directives on Standardizing the Use of SCI-Paper-Related Indicators in Higher Education and Establishing a Correct Evaluation Orientation (Jiao Ji Ke [2020] No. 2). (In Chinese.) https://hudong.moe.gov.cn/srcsite/A16/moe_784/202002/t20200223_423334.html

National Science Board. (2025). Discovery: R&D Activity and Research Publications. *Science and Engineering Indicators 2026*. NSB-2025-7. https://ncses.nsf.gov/pubs/nsb20257/

Open Science Collaboration. (2015). Estimating the Reproducibility of Psychological Science. *Science*, 349(6251), aac4716. https://doi.org/10.1126/science.aac4716

Park, M., Leahey, E., & Funk, R. J. (2023). Papers and Patents Are Becoming Less Disruptive over Time. *Nature*, 613. https://doi.org/10.1038/s41586-022-05543-x

Park, M., Leahey, E., & Funk, R. J. (2026). Reply to: Dataset Artefacts Can Partially Drive the Measured Decline in Disruption. *Nature*, 656(8127). https://doi.org/10.1038/s41586-026-10788-x

Peng, D., et al. (2025). Monitoring of Global OA Journals and Article Processing Charges in 2024 and Suggestions for China's Response Strategies. *Chinese Journal of Scientific and Technical Periodicals*, 36(11). (In Chinese.) https://doi.org/10.11946/cjstp.202510161239

Posner, R. A. (2014). *Economic Analysis of Law*. Wolters Kluwer.

Power, M. (1997). *The Audit Society: Rituals of Verification*. Oxford University Press.

Redner, H. (2023). *The Humanities, the Social Sciences and the University: A Study in Knowledge Production*. Routledge.

Roncaglia, A. (2019). *The Age of Fragmentation: A History of Contemporary Economic Thought*. Cambridge: Cambridge University Press.

Rosa, H. (2013). *Social Acceleration: A New Theory of Modernity*. New York: Columbia University Press. Chinese edition: Dong Lu, trans., *Acceleration: Changes in the Structure of Time in Modern Society*. Beijing: Peking University Press, 2015. (In Chinese.)

Schwabe, U., Peter, F., & Buchholz, S. (2024). Tenure Track Career Options to Full Professorship in German Academia: Recent Developments and Challenges. In E. Pekkola & T. Siekkinen (Eds.), *Tenure Tracks in European Universities: Managing Careers in Academia*. Cheltenham: Edward Elgar Publishing.

Scott, J. C. (1998). *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. New Haven: Yale University Press.

Sever, R., et al. (2019). bioRxiv: The Preprint Server for Biology. *bioRxiv*. https://doi.org/10.1101/833400 [Preprint, not yet peer-reviewed]

Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Shapiro, F. R. (Ed.). (2006). *The Yale Book of Quotations*. New Haven: Yale University Press.

Shiller, R. J. (2019). *Narrative Economics: How Stories Go Viral and Drive Major Economic Events*. Princeton, NJ: Princeton University Press.

Shore, C., & Wright, S. (2024). *Audit Culture: How Indicators and Rankings Are Reshaping the World*. Pluto Press.

Sinatra, R., et al. (2016). Quantifying the Evolution of Individual Scientific Impact. *Science*, 354(6312), aaf5239. https://doi.org/10.1126/science.aaf5239

Slaughter, S., & Rhoades, G. (2004). *Academic Capitalism and the New Economy*. Johns Hopkins University Press.

Smith, R. (2006). Peer Review: A Flawed Process at the Heart of Science and Journals. *Journal of the Royal Society of Medicine*, 99(4). https://doi.org/10.1258/jrsm.99.4.178

Sowell, T. (1980). *Knowledge and Decisions*. New York: Basic Books.

Stack, M. (2020). Academic Stars and University Rankings in Higher Education: Impacts on Policy and Practice. *Policy Reviews in Higher Education*, 4(1). https://doi.org/10.1080/23322969.2019.1667859

Steenrod, N. E. (1951). *The Topology of Fibre Bundles*. Princeton, NJ: Princeton University Press.

Stern, A. M., et al. (2014). Financial Costs and Personal Consequences of Research Misconduct Resulting in Retracted Publications. *eLife*, 3, e02956. https://doi.org/10.7554/eLife.02956

Strathern, M. (1997). 'Improving Ratings': Audit in the British University System. *European Review*, 5(3). https://doi.org/10.1002/(SICI)1234-981X(199707)5:3%3C305::AID-EURO184%3E3.0.CO;2-4

Stroebe, W., Postmes, T., & Spears, R. (2012). Scientific Misconduct and the Myth of Self-Correction in Science. *Perspectives on Psychological Science*, 7(6). https://doi.org/10.1177/1745691612460687

Tian, M., Su, Y., & Ru, X. (2016). Perish or Publish in China: Pressures on Young Chinese Scholars to Publish in Internationally Indexed Journals. *Publications*, 4(2), 9. https://doi.org/10.3390/publications4020009

Tiles, J. E., & Oberdiek, H. (1995). *Living in a Technological Culture: Human Tools and Human Values*. London: Routledge.

Tishby, N., Pereira, F. C., & Bialek, W. (1999). The Information Bottleneck Method. In *Proceedings of the 37th Allerton Conference on Communication, Control and Computing*.

Uslu, B. (2020). A Path for Ranking Success: What Does the Expanded Indicator-Set of International University Rankings Suggest? *Higher Education*, 80(5). https://doi.org/10.1007/s10734-020-00527-0

von Neumann, J. (1947). The Mathematician. In *The Works of the Mind*. University of Chicago Press.

von Neumann, J. (1955). Can We Survive Technology? *Fortune*.

Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. New York: Springer.

Weizenbaum, J. (1976). *Computer Power and Human Reason: From Judgment to Calculation*. San Francisco: W. H. Freeman.

Whitley, R. (2010). Reconfiguring the Public Sciences: The Impact of Governance Changes on Authority and Innovation in Public Science Systems. In R. Whitley, J. Gläser, & L. Engwall (Eds.), *Reconfiguring Knowledge Production: Changing Authority Relationships in the Sciences and Their Consequences for Intellectual Innovation*. Oxford: Oxford University Press.

Wu, L., Wang, D., & Evans, J. A. (2019). Large Teams Develop and Small Teams Disrupt Science and Technology. *Nature*, 566. https://doi.org/10.1038/s41586-019-0941-9

Zheng, J., & Meister, M. (2025). The Unbearable Slowness of Being: Why Do We Live at 10 bits/s? *Neuron*, 113(2). https://doi.org/10.1016/j.neuron.2024.11.008

Zhao, Dingxin (2004). The Explanatory Tradition or the Interpretive Tradition? Where Lies the Way Out for Contemporary Humanities and Social Sciences. *Social Observation*, (6). (In Chinese.)

Zhao, Dingxin (2019). Time, Temporality, and Wisdom: The Essence of Historical Sociology. *Sociological Review*, 7(1). (In Chinese.)

Zhao, Dingxin (2022). The Levels of Explanation and the Hermeneutic Circle. In Y. Liang (Ed.), *Anthropological Studies* (Vols. 15–16). Beijing: The Commercial Press. (In Chinese.)